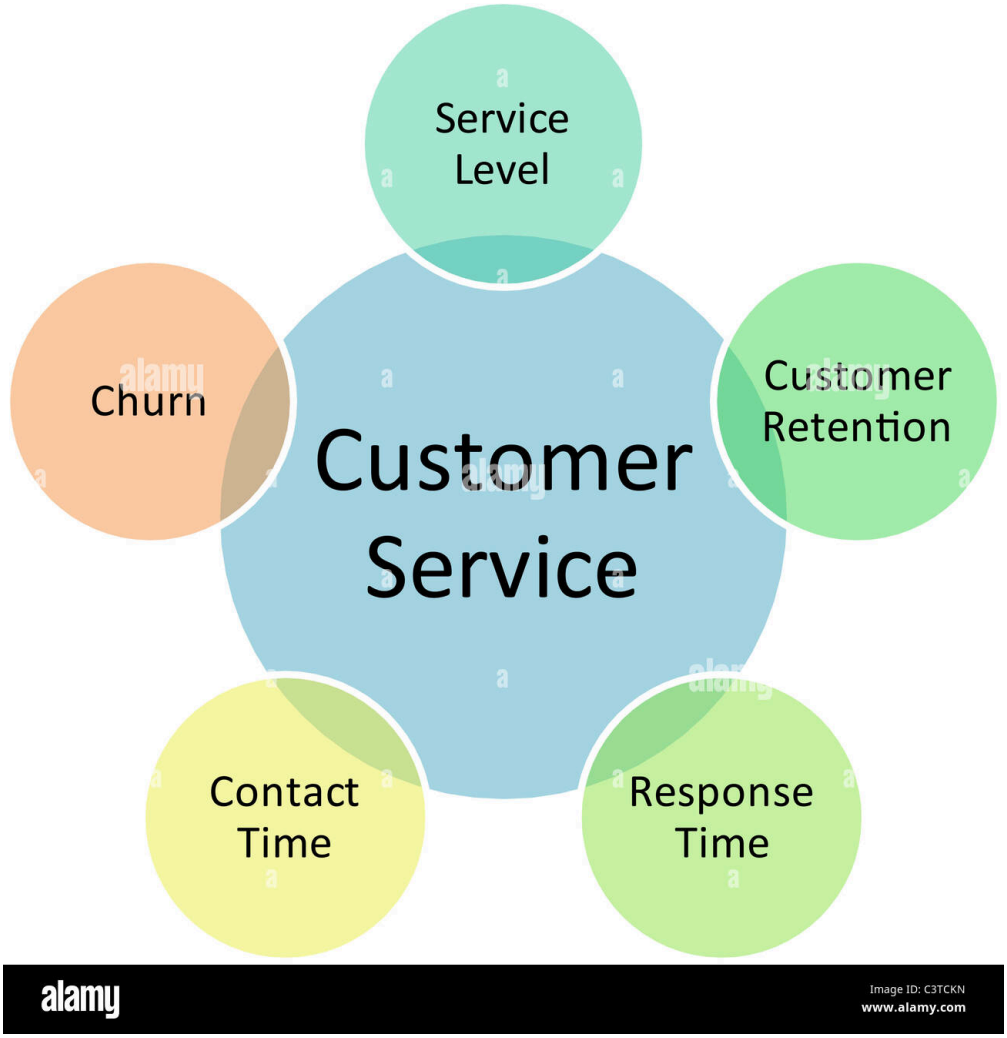
CUSTOMER SERVICE REQUEST ANALYSIS



Creating a Problem Statement

1.Find the shape of the dataset (rows & columns).

2.Identify columns with missing (null) values.

3.Remove records where Closed Date is missing.

4.Convert Created Date and Closed Date into datetime format.

5.Remove records where Closed Date is earlier than Created Date.

6.Fill missing values in the City column with "Unknown City".

7.Find the total number of complaints for each Complaint Type.

8.Find the number of complaints in each city.

9.Create a table showing complaint types across different cities.

10.Plot a bar chart of complaint types. Plot complaints distribution across cities.

11.Create a stacked bar chart of complaint types by city.

12.Calculate average response time for each complaint type.

13.Compare response time across complaint types using a bar chart.

14.Perform Kruskal-Wallis test on response time by complaint type.

15.Check p-value and conclude: Reject or Accept Null Hypothesis answer

Importing required libraries

In [1]:
```
import numpy as np
import pandas as pd
import matplotlib.pyplot as plt
from scipy.stats import kruskal
```

Importing the dataset

In [2]:
```
df = pd.read_csv("311_Service_Requests_from_2010_to_Present.csv", chunksize=1000

for chunk in df:
    print(chunk.head())
```

```
   Unique Key            Created Date             Closed Date Agency  \
0    32310363  12/31/2015 11:59:45 PM  01/01/2016 12:55:15 AM   NYPD
1    32309934  12/31/2015 11:59:44 PM  01/01/2016 01:26:57 AM   NYPD
2    32309159  12/31/2015 11:59:29 PM  01/01/2016 04:51:03 AM   NYPD
3    32305098  12/31/2015 11:57:46 PM  01/01/2016 07:43:13 AM   NYPD
4    32306529  12/31/2015 11:56:58 PM  01/01/2016 03:24:42 AM   NYPD

                      Agency Name           Complaint Type  \
0  New York City Police Department  Noise - Street/Sidewalk
1  New York City Police Department         Blocked Driveway
2  New York City Police Department         Blocked Driveway
3  New York City Police Department          Illegal Parking
4  New York City Police Department          Illegal Parking

                     Descriptor    Location Type  Incident Zip  \
0             Loud Music/Party  Street/Sidewalk       10034.0
1                    No Access  Street/Sidewalk       11105.0
2                    No Access  Street/Sidewalk       10458.0
3  Commercial Overnight Parking  Street/Sidewalk       10461.0
4             Blocked Sidewalk  Street/Sidewalk       11373.0

        Incident Address  ... Bridge Highway Name Bridge Highway Direction  \
0    71 VERMILYEA AVENUE  ...                 NaN                      NaN
1        27-07 23 AVENUE  ...                 NaN                      NaN
2  2897 VALENTINE AVENUE  ...                 NaN                      NaN
3    2940 BAISLEY AVENUE  ...                 NaN                      NaN
4         87-14 57 ROAD  ...                 NaN                      NaN

  Road Ramp Bridge Highway Segment Garage Lot Name Ferry Direction  \
0       NaN                    NaN             NaN             NaN
1       NaN                    NaN             NaN             NaN
2       NaN                    NaN             NaN             NaN
3       NaN                    NaN             NaN             NaN
4       NaN                    NaN             NaN             NaN

  Ferry Terminal Name   Latitude  Longitude  \
0                 NaN  40.865682 -73.923501
1                 NaN  40.775945 -73.915094
2                 NaN  40.870325 -73.888525
3                 NaN  40.835994 -73.828379
4                 NaN  40.733060 -73.874170

                                 Location
0   (40.86568153633767, -73.92350095571744)
1  (40.775945312321085, -73.91509393898605)
2  (40.870324522111424, -73.88852464418646)
3   (40.83599404683083, -73.82837939584206)
4  (40.733059618956815, -73.87416975810375)

[5 rows x 53 columns]
       Unique Key            Created Date             Closed Date Agency  \
10000    32242277  12/20/2015 05:20:53 PM  12/20/2015 06:34:32 PM   NYPD
10001    32239628  12/20/2015 05:11:34 PM  12/20/2015 07:33:28 PM   NYPD
10002    32241437  12/20/2015 05:11:01 PM  12/20/2015 10:45:26 PM   NYPD
10003    32239675  12/20/2015 05:10:27 PM  12/20/2015 05:27:02 PM   NYPD
10004    32242151  12/20/2015 05:09:36 PM  12/20/2015 05:52:07 PM   NYPD

                         Agency Name    Complaint Type  \
10000  New York City Police Department      Animal Abuse
10001  New York City Police Department  Blocked Driveway
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```
10002  New York City Police Department   Illegal Parking
10003  New York City Police Department           Traffic
10004  New York City Police Department  Blocked Driveway

                     Descriptor               Location Type  Incident Zip  \
10000  Other (complaint details)  Residential Building/House       10011.0
10001            Partial Access             Street/Sidewalk       11235.0
10002           Blocked Hydrant             Street/Sidewalk       11372.0
10003      Truck Route Violation             Street/Sidewalk       11218.0
10004                 No Access             Street/Sidewalk       11355.0

             Incident Address  ... Bridge Highway Name  \
10000       130 WEST 22 STREET  ...                 NaN
10001  2920 BRIGHTON 12 STREET  ...                 NaN
10002                      NaN  ...                 NaN
10003                      NaN  ...                 NaN
10004         141-12 45 AVENUE  ...                 NaN

      Bridge Highway Direction Road Ramp Bridge Highway Segment  \
10000                      NaN       NaN                    NaN
10001                      NaN       NaN                    NaN
10002                      NaN       NaN                    NaN
10003                      NaN       NaN                    NaN
10004                      NaN       NaN                    NaN

      Garage Lot Name Ferry Direction Ferry Terminal Name   Latitude  \
10000             NaN             NaN                 NaN  40.742663
10001             NaN             NaN                 NaN  40.581397
10002             NaN             NaN                 NaN  40.755755
10003             NaN             NaN                 NaN  40.641746
10004             NaN             NaN                 NaN  40.753035

       Longitude                                 Location
10000 -73.994331   (40.74266349678957, -73.99433063580953)
10001 -73.955611   (40.58139659782926, -73.95561141256746)
10002 -73.883508   (40.75575482173939, -73.88350790971228)
10003 -73.985126  (40.641745763760696, -73.98512556325805)
10004 -73.819610   (40.753034774757396, -73.8196095047382)

[5 rows x 53 columns]
       Unique Key            Created Date             Closed Date Agency  \
20000    32183984  12/11/2015 10:52:59 AM  12/11/2015 01:15:02 PM   NYPD
20001    32180235  12/11/2015 10:52:12 AM  12/11/2015 01:36:16 PM   NYPD
20002    32182626  12/11/2015 10:49:52 AM  12/11/2015 01:02:10 PM   NYPD
20003    32179684  12/11/2015 10:48:58 AM  12/12/2015 12:42:07 AM   NYPD
20004    32181783  12/11/2015 10:46:08 AM  12/11/2015 01:02:12 PM   NYPD

                         Agency Name       Complaint Type  \
20000  New York City Police Department             Drinking
20001  New York City Police Department     Blocked Driveway
20002  New York City Police Department      Illegal Parking
20003  New York City Police Department  Homeless Encampment
20004  New York City Police Department     Blocked Driveway

                         Descriptor        Location Type  Incident Zip  \
20000      After Hours - Licensed Est  Club/Bar/Restaurant       11221.0
20001                 Partial Access      Street/Sidewalk       11221.0
20002  Double Parked Blocking Vehicle      Street/Sidewalk       10463.0
20003                            NaN      Street/Sidewalk       11105.0
20004                 Partial Access      Street/Sidewalk       10463.0
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              Incident Address  ... Bridge Highway Name  \
20000       1152 DEKALB AVENUE  ...                 NaN
20001     1025 BUSHWICK AVENUE  ...                 NaN
20002                      NaN  ...                 NaN
20003  19-29 DITMARS BOULEVARD  ...                 NaN
20004        2676 HEATH AVENUE  ...                 NaN

      Bridge Highway Direction Road Ramp Bridge Highway Segment  \
20000                      NaN       NaN                    NaN
20001                      NaN       NaN                    NaN
20002                      NaN       NaN                    NaN
20003                      NaN       NaN                    NaN
20004                      NaN       NaN                    NaN

      Garage Lot Name Ferry Direction Ferry Terminal Name   Latitude  \
20000             NaN             NaN                 NaN  40.694608
20001             NaN             NaN                 NaN  40.691431
20002             NaN             NaN                 NaN  40.884256
20003             NaN             NaN                 NaN  40.780687
20004             NaN             NaN                 NaN  40.870708

       Longitude                                 Location
20000 -73.928980   (40.69460843160586, -73.92898026982887)
20001 -73.921526    (40.69143063658496, -73.9215264676878)
20002 -73.887586  (40.884255673730884, -73.88758632273493)
20003 -73.917139   (40.78068696247032, -73.91713884070063)
20004 -73.905800   (40.87070770667293, -73.90579987814343)

[5 rows x 53 columns]
       Unique Key            Created Date             Closed Date Agency  \
30000    32108998  12/01/2015 05:00:33 PM  12/01/2015 06:46:47 PM   NYPD
30001    32104940  12/01/2015 04:59:08 PM  12/01/2015 06:46:46 PM   NYPD
30002    32105016  12/01/2015 04:57:05 PM  12/01/2015 07:14:06 PM   NYPD
30003    32108969  12/01/2015 04:56:27 PM  12/01/2015 06:22:26 PM   NYPD
30004    32104894  12/01/2015 04:55:09 PM  12/01/2015 06:27:42 PM   NYPD

                         Agency Name           Complaint Type  \
30000  New York City Police Department         Derelict Vehicle
30001  New York City Police Department         Derelict Vehicle
30002  New York City Police Department  Noise - Street/Sidewalk
30003  New York City Police Department                  Traffic
30004  New York City Police Department          Illegal Parking

                        Descriptor    Location Type  Incident Zip  \
30000             With License Plate  Street/Sidewalk       11419.0
30001             With License Plate  Street/Sidewalk       11419.0
30002                   Loud Talking  Street/Sidewalk       10027.0
30003            Congestion/Gridlock  Street/Sidewalk       10013.0
30004  Posted Parking Sign Violation  Street/Sidewalk       10031.0

        Incident Address  ... Bridge Highway Name Bridge Highway Direction  \
30000  107-37 122 STREET  ...                 NaN                      NaN
30001  107-43 122 STREET  ...                 NaN                      NaN
30002   260 LENOX AVENUE  ...                 NaN                      NaN
30003               NaN  ...                 NaN                      NaN
30004               NaN  ...                 NaN                      NaN

      Road Ramp Bridge Highway Segment Garage Lot Name Ferry Direction  \
30000       NaN                    NaN             NaN             NaN
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30001       NaN                    NaN             NaN             NaN
30002       NaN                    NaN             NaN             NaN
30003       NaN                    NaN             NaN             NaN
30004       NaN                    NaN             NaN             NaN

      Ferry Terminal Name   Latitude  Longitude  \
30000                 NaN  40.684186 -73.820113
30001                 NaN  40.684068 -73.820056
30002                 NaN  40.806639 -73.946301
30003                 NaN  40.723308 -74.002980
30004                 NaN  40.818394 -73.950879

                                     Location
30000   (40.68418587059898, -73.8201130468493)
30001  (40.68406775546039, -73.82005567591126)
30002  (40.80663919669058, -73.94630058482072)
30003   (40.7233075908167, -74.00297997118642)
30004   (40.8183941375894, -73.95087944274451)

[5 rows x 53 columns]
       Unique Key            Created Date             Closed Date Agency  \
40000    32028842  11/21/2015 12:37:55 AM  11/22/2015 08:16:28 AM   NYPD
40001    32029152  11/21/2015 12:37:54 AM  11/21/2015 05:40:29 AM   NYPD
40002    32034751  11/21/2015 12:37:27 AM  11/21/2015 05:15:44 AM   NYPD
40003    32033670  11/21/2015 12:37:06 AM  11/21/2015 03:40:52 AM   NYPD
40004    32030075  11/21/2015 12:36:40 AM  11/21/2015 02:39:12 AM   NYPD

                          Agency Name           Complaint Type  \
40000  New York City Police Department  Noise - Street/Sidewalk
40001  New York City Police Department          Illegal Parking
40002  New York City Police Department       Noise - Commercial
40003  New York City Police Department  Noise - Street/Sidewalk
40004  New York City Police Department          Illegal Parking

                          Descriptor        Location Type  Incident Zip  \
40000                    Loud Talking      Street/Sidewalk           NaN
40001  Double Parked Blocking Vehicle      Street/Sidewalk       11102.0
40002               Loud Music/Party  Club/Bar/Restaurant       11215.0
40003               Loud Music/Party      Street/Sidewalk       11434.0
40004  Double Parked Blocking Vehicle      Street/Sidewalk       10034.0

             Incident Address  ... Bridge Highway Name  \
40000  48 DR BILLY TAYLOR WAY  ...                 NaN
40001          26-45 9 STREET  ...                 NaN
40002  290 PROSPECT PARK WEST  ...                 NaN
40003       172-40 133 AVENUE  ...                 NaN
40004     565 WEST 207 STREET  ...                 NaN

      Bridge Highway Direction Road Ramp Bridge Highway Segment  \
40000                      NaN       NaN                    NaN
40001                      NaN       NaN                    NaN
40002                      NaN       NaN                    NaN
40003                      NaN       NaN                    NaN
40004                      NaN       NaN                    NaN

      Garage Lot Name Ferry Direction Ferry Terminal Name   Latitude  \
40000             NaN             NaN                 NaN        NaN
40001             NaN             NaN                 NaN  40.774773
40002             NaN             NaN                 NaN  40.657802
40003             NaN             NaN                 NaN  40.676016
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40004             NaN             NaN                 NaN  40.866349

       Longitude                                  Location
40000        NaN                                       NaN
40001 -73.930621  (40.77477272473301, -73.93062059642939)
40002 -73.982772  (40.65780241585331, -73.98277206373291)
40003 -73.768707  (40.67601613067663, -73.76870740383659)
40004 -73.920337  (40.86634910127423, -73.92033664725604)

[5 rows x 53 columns]
       Unique Key            Created Date             Closed Date Agency  \
50000    31959891  11/11/2015 02:57:21 PM  11/11/2015 05:15:48 PM   NYPD
50001    31963670  11/11/2015 02:54:58 PM  11/11/2015 07:33:31 PM   NYPD
50002    31960197  11/11/2015 02:52:42 PM  11/11/2015 07:32:23 PM   NYPD
50003    31962017  11/11/2015 02:52:20 PM  11/11/2015 09:59:51 PM   NYPD
50004    31961600  11/11/2015 02:50:53 PM  11/11/2015 09:40:46 PM   NYPD

                          Agency Name           Complaint Type  \
50000  New York City Police Department  Noise - Street/Sidewalk
50001  New York City Police Department         Blocked Driveway
50002  New York City Police Department         Blocked Driveway
50003  New York City Police Department       Noise - Commercial
50004  New York City Police Department         Blocked Driveway

             Descriptor     Location Type  Incident Zip  \
50000  Loud Music/Party   Street/Sidewalk       11372.0
50001         No Access   Street/Sidewalk       11105.0
50002         No Access   Street/Sidewalk       11207.0
50003  Loud Music/Party  Store/Commercial       11372.0
50004    Partial Access   Street/Sidewalk       11373.0

             Incident Address  ... Bridge Highway Name  \
50000  80-01 ROOSEVELT AVENUE  ...                 NaN
50001         23-38 33 STREET  ...                 NaN
50002   708 PENNSYLVAN AVENUE  ...                 NaN
50003  80-12 ROOSEVELT AVENUE  ...                 NaN
50004           86-32 55 ROAD  ...                 NaN

      Bridge Highway Direction Road Ramp Bridge Highway Segment  \
50000                      NaN       NaN                    NaN
50001                      NaN       NaN                    NaN
50002                      NaN       NaN                    NaN
50003                      NaN       NaN                    NaN
50004                      NaN       NaN                    NaN

      Garage Lot Name Ferry Direction Ferry Terminal Name   Latitude  \
50000             NaN             NaN                 NaN  40.747462
50001             NaN             NaN                 NaN  40.772828
50002             NaN             NaN                 NaN  40.660369
50003             NaN             NaN                 NaN  40.747454
50004             NaN             NaN                 NaN  40.733790

       Longitude                                  Location
50000 -73.885652  (40.74746233713562, -73.88565173450539)
50001 -73.912737   (40.77282827232839, -73.9127367129827)
50002 -73.891997  (40.66036877789996, -73.89199684138318)
50003 -73.885569  (40.74745402045867, -73.88556874122851)
50004 -73.877199  (40.73379025008941, -73.87719933735931)

[5 rows x 53 columns]
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       Unique Key            Created Date             Closed Date Agency  \
60000    31894826  11/02/2015 06:45:17 AM  11/02/2015 08:52:00 AM   NYPD
60001    31898919  11/02/2015 06:42:58 AM  11/02/2015 09:45:08 AM   NYPD
60002    31897233  11/02/2015 06:42:19 AM  11/02/2015 11:01:50 AM   NYPD
60003    31900515  11/02/2015 06:42:17 AM  11/02/2015 07:53:39 AM   NYPD
60004    31892764  11/02/2015 06:41:56 AM  11/02/2015 07:05:30 PM   NYPD

                          Agency Name    Complaint Type          Descriptor  \
60000  New York City Police Department   Illegal Parking     Blocked Hydrant
60001  New York City Police Department  Derelict Vehicle  With License Plate
60002  New York City Police Department  Blocked Driveway           No Access
60003  New York City Police Department   Noise - Vehicle       Engine Idling
60004  New York City Police Department   Illegal Parking     Blocked Hydrant

         Location Type  Incident Zip               Incident Address  ...  \
60000  Street/Sidewalk       11223.0                152 LAKE STREET  ...
60001  Street/Sidewalk       11354.0  40-05 COLLEGE POINT BOULEVARD  ...
60002  Street/Sidewalk       11433.0              104-29 171 STREET  ...
60003  Street/Sidewalk       10027.0            259 WEST 123 STREET  ...
60004  Street/Sidewalk       11218.0              405 EAST 7 STREET  ...

      Bridge Highway Name Bridge Highway Direction Road Ramp  \
60000                 NaN                      NaN       NaN
60001                 NaN                      NaN       NaN
60002                 NaN                      NaN       NaN
60003                 NaN                      NaN       NaN
60004                 NaN                      NaN       NaN

      Bridge Highway Segment Garage Lot Name Ferry Direction  \
60000                    NaN             NaN             NaN
60001                    NaN             NaN             NaN
60002                    NaN             NaN             NaN
60003                    NaN             NaN             NaN
60004                    NaN             NaN             NaN

      Ferry Terminal Name   Latitude  Longitude  \
60000                 NaN  40.600462 -73.973593
60001                 NaN  40.758049 -73.834215
60002                 NaN  40.702231 -73.786696
60003                 NaN  40.808362 -73.950909
60004                 NaN  40.641977 -73.972431

                                     Location
60000  (40.600462028437086, -73.97359331316606)
60001    (40.75804942215372, -73.8342148123826)
60002  (40.702230581537705, -73.78669626113849)
60003    (40.8083621929931, -73.95090852035001)
60004   (40.64197672784396, -73.97243102936937)

[5 rows x 53 columns]
       Unique Key            Created Date             Closed Date Agency  \
70000    31839143  10/23/2015 07:27:17 PM  10/23/2015 08:35:57 PM   NYPD
70001    31842871  10/23/2015 07:23:05 PM  10/24/2015 07:35:51 AM   NYPD
70002    31841770  10/23/2015 07:23:05 PM  10/23/2015 08:02:04 PM   NYPD
70003    31846521  10/23/2015 07:22:06 PM  10/25/2015 04:01:22 AM   NYPD
70004    31839855  10/23/2015 07:21:35 PM  10/23/2015 11:12:21 PM   NYPD

                          Agency Name           Complaint Type  \
70000  New York City Police Department         Blocked Driveway
70001  New York City Police Department  Noise - Street/Sidewalk
```

```
70002  New York City Police Department       Noise - Commercial
70003  New York City Police Department         Blocked Driveway
70004  New York City Police Department         Derelict Vehicle

               Descriptor     Location Type  Incident Zip  \
70000           No Access   Street/Sidewalk       11249.0
70001    Loud Music/Party   Street/Sidewalk       11226.0
70002    Loud Music/Party  Store/Commercial       10025.0
70003           No Access   Street/Sidewalk       10466.0
70004  With License Plate   Street/Sidewalk       11357.0

             Incident Address  ... Bridge Highway Name  \
70000      115 SOUTH 1 STREET  ...                 NaN
70001     175 CLARKSON AVENUE  ...                 NaN
70002     924 COLUMBUS AVENUE  ...                 NaN
70003  4015 BAYCHESTER AVENUE  ...                 NaN
70004               18 AVENUE  ...                 NaN

      Bridge Highway Direction Road Ramp Bridge Highway Segment  \
70000                      NaN       NaN                    NaN
70001                      NaN       NaN                    NaN
70002                      NaN       NaN                    NaN
70003                      NaN       NaN                    NaN
70004                      NaN       NaN                    NaN

      Garage Lot Name Ferry Direction Ferry Terminal Name   Latitude  \
70000             NaN             NaN                 NaN  40.714318
70001             NaN             NaN                 NaN  40.655184
70002             NaN             NaN                 NaN  40.798785
70003             NaN             NaN                 NaN  40.890873
70004             NaN             NaN                 NaN  40.783189

       Longitude                                 Location
70000 -73.962802  (40.714318021793936, -73.96280227689175)
70001 -73.954164   (40.65518428643387, -73.95416395328084)
70002 -73.963214    (40.7987849389821, -73.96321377679931)
70003 -73.842889   (40.89087308305596, -73.84288942514577)
70004 -73.820091   (40.78318930936784, -73.82009111994392)

[5 rows x 53 columns]
       Unique Key            Created Date             Closed Date Agency  \
80000    31751346  10/14/2015 02:18:51 PM  10/14/2015 03:32:24 PM   NYPD
80001    31751306  10/14/2015 02:18:32 PM  10/14/2015 03:24:17 PM   NYPD
80002    31750286  10/14/2015 02:15:53 PM  10/15/2015 10:36:32 AM   NYPD
80003    31748347  10/14/2015 02:14:03 PM  10/15/2015 04:43:52 AM   NYPD
80004    31754311  10/14/2015 02:11:06 PM  10/14/2015 04:16:39 PM   NYPD

                          Agency Name           Complaint Type  \
80000  New York City Police Department          Illegal Parking
80001  New York City Police Department  Noise - Street/Sidewalk
80002  New York City Police Department         Derelict Vehicle
80003  New York City Police Department          Illegal Parking
80004  New York City Police Department         Derelict Vehicle

                         Descriptor    Location Type  Incident Zip  \
80000  Posted Parking Sign Violation  Street/Sidewalk       10306.0
80001                   Loud Talking  Street/Sidewalk       10306.0
80002             With License Plate  Street/Sidewalk       11417.0
80003               Blocked Sidewalk  Street/Sidewalk       11101.0
80004             With License Plate  Street/Sidewalk       10469.0
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           Incident Address  ... Bridge Highway Name Bridge Highway Direction  \
80000   200 ADELAIDE AVENUE  ...                 NaN                      NaN
80001  750 PATTERSON AVENUE  ...                 NaN                      NaN
80002                   NaN  ...                 NaN                      NaN
80003                   NaN  ...                 NaN                      NaN
80004      1168 MACE AVENUE  ...                 NaN                      NaN

      Road Ramp Bridge Highway Segment Garage Lot Name Ferry Direction  \
80000       NaN                    NaN             NaN             NaN
80001       NaN                    NaN             NaN             NaN
80002       NaN                    NaN             NaN             NaN
80003       NaN                    NaN             NaN             NaN
80004       NaN                    NaN             NaN             NaN

      Ferry Terminal Name   Latitude  Longitude  \
80000                 NaN  40.561690 -74.124622
80001                 NaN  40.573191 -74.089357
80002                 NaN  40.674723 -73.858381
80003                 NaN  40.743537 -73.950711
80004                 NaN  40.862943 -73.853169

                                   Location
80000   (40.5616902523158, -74.12462211525013)
80001  (40.57319090616437, -74.08935661571287)
80002  (40.67472318987006, -73.85838137799163)
80003  (40.743536899151906, -73.9507108878168)
80004   (40.862942878320005, -73.853168613916)

[5 rows x 53 columns]
       Unique Key            Created Date             Closed Date Agency  \
90000    31677749  10/05/2015 12:04:50 AM  10/05/2015 01:35:15 AM   NYPD
90001    31681173  10/05/2015 12:04:39 AM  10/05/2015 05:28:56 AM   NYPD
90002    31682299  10/05/2015 12:03:33 AM  10/05/2015 04:22:19 AM   NYPD
90003    31678981  10/05/2015 12:02:35 AM  10/05/2015 05:28:56 AM   NYPD
90004    31679926  10/05/2015 12:01:54 AM  10/05/2015 12:45:46 AM   NYPD

                         Agency Name           Complaint Type  \
90000  New York City Police Department       Noise - Commercial
90001  New York City Police Department       Noise - Commercial
90002  New York City Police Department  Noise - Street/Sidewalk
90003  New York City Police Department       Noise - Commercial
90004  New York City Police Department  Noise - Street/Sidewalk

             Descriptor        Location Type  Incident Zip  \
90000  Loud Music/Party  Club/Bar/Restaurant       11237.0
90001  Loud Music/Party  Club/Bar/Restaurant       10034.0
90002      Loud Talking      Street/Sidewalk       10027.0
90003  Loud Music/Party  Club/Bar/Restaurant       10040.0
90004  Loud Music/Party      Street/Sidewalk       10025.0

           Incident Address  ... Bridge Highway Name Bridge Highway Direction  \
90000   1495 DEKALB AVENUE  ...                 NaN                      NaN
90001       3795 10 AVENUE  ...                 NaN                      NaN
90002  118 WEST 123 STREET  ...                 NaN                      NaN
90003    65 SHERMAN AVENUE  ...                 NaN                      NaN
90004  141 WEST 109 STREET  ...                 NaN                      NaN

      Road Ramp Bridge Highway Segment Garage Lot Name Ferry Direction  \
90000       NaN                    NaN             NaN             NaN
```

```
90001       NaN                    NaN             NaN             NaN
90002       NaN                    NaN             NaN             NaN
90003       NaN                    NaN             NaN             NaN
90004       NaN                    NaN             NaN             NaN

      Ferry Terminal Name   Latitude  Longitude  \
90000                 NaN  40.701696 -73.921814
90001                 NaN  40.862032 -73.920689
90002                 NaN  40.806810 -73.947258
90003                 NaN  40.863282 -73.926783
90004                 NaN  40.801702 -73.962692

                                     Location
90000   (40.70169626675707, -73.92181372712216)
90001     (40.8620319333462, -73.9206888817376)
90002   (40.80680981145933, -73.94725769076324)
90003  (40.863282054634674, -73.92678280024691)
90004  (40.801702416194175, -73.96269204190232)

[5 rows x 53 columns]
        Unique Key            Created Date             Closed Date Agency  \
100000    31612425  09/25/2015 10:11:52 PM  09/26/2015 04:09:41 AM   NYPD
100001    31614483  09/25/2015 10:11:42 PM  09/25/2015 11:04:15 PM   NYPD
100002    31612149  09/25/2015 10:11:06 PM  09/25/2015 11:04:14 PM   NYPD
100003    31612487  09/25/2015 10:09:32 PM  09/26/2015 03:54:58 AM   NYPD
100004    31612611  09/25/2015 10:09:11 PM  09/25/2015 10:57:29 PM   NYPD

                           Agency Name           Complaint Type  \
100000  New York City Police Department         Blocked Driveway
100001  New York City Police Department  Noise - Street/Sidewalk
100002  New York City Police Department  Noise - Street/Sidewalk
100003  New York City Police Department       Noise - Commercial
100004  New York City Police Department          Illegal Parking

              Descriptor        Location Type  Incident Zip  \
100000    Partial Access      Street/Sidewalk       10460.0
100001      Loud Talking      Street/Sidewalk       10001.0
100002      Loud Talking      Street/Sidewalk       10001.0
100003  Loud Music/Party  Club/Bar/Restaurant       11105.0
100004   Blocked Hydrant      Street/Sidewalk       11228.0

                Incident Address  ... Bridge Highway Name  \
100000  1243 EAST TREMONT AVENUE  ...                 NaN
100001        124 WEST 34 STREET  ...                 NaN
100002        114 WEST 34 STREET  ...                 NaN
100003   38-01 DITMARS BOULEVARD  ...                 NaN
100004            1474 76 STREET  ...                 NaN

       Bridge Highway Direction Road Ramp Bridge Highway Segment  \
100000                      NaN       NaN                    NaN
100001                      NaN       NaN                    NaN
100002                      NaN       NaN                    NaN
100003                      NaN       NaN                    NaN
100004                      NaN       NaN                    NaN

       Garage Lot Name Ferry Direction Ferry Terminal Name   Latitude  \
100000             NaN             NaN                 NaN  40.839703
100001             NaN             NaN                 NaN  40.750214
100002             NaN             NaN                 NaN  40.750132
100003             NaN             NaN                 NaN  40.773296
```

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100004             NaN             NaN                 NaN  40.617461

        Longitude                                 Location
100000 -73.871612  (40.839703487486965, -73.87161170359985)
100001 -73.988808    (40.75021389729075, -73.9888079624495)
100002 -73.988617   (40.75013153611778, -73.98861669048847)
100003 -73.906674    (40.7732956167602, -73.90667419984779)
100004 -74.004366   (40.61746074949296, -74.00436561493296)

[5 rows x 53 columns]
        Unique Key            Created Date             Closed Date Agency  \
110000    31558545  09/17/2015 10:58:39 PM  09/18/2015 02:04:37 AM   NYPD
110001    31553660  09/17/2015 10:57:33 PM  09/18/2015 03:06:59 AM   NYPD
110002    31552514  09/17/2015 10:57:33 PM  09/18/2015 02:14:28 AM   NYPD
110003    31557857  09/17/2015 10:57:10 PM  09/18/2015 07:53:34 AM   NYPD
110004    31554162  09/17/2015 10:55:20 PM  09/18/2015 05:21:50 AM   NYPD

                           Agency Name           Complaint Type  \
110000  New York City Police Department       Noise - Commercial
110001  New York City Police Department         Blocked Driveway
110002  New York City Police Department  Noise - Street/Sidewalk
110003  New York City Police Department         Blocked Driveway
110004  New York City Police Department          Illegal Parking

              Descriptor        Location Type  Incident Zip  \
110000      Loud Talking  Club/Bar/Restaurant       11217.0
110001         No Access      Street/Sidewalk       11419.0
110002  Loud Music/Party      Street/Sidewalk       10037.0
110003    Partial Access      Street/Sidewalk       11236.0
110004   Blocked Hydrant      Street/Sidewalk       11415.0

          Incident Address  ... Bridge Highway Name Bridge Highway Direction  \
110000   84 ST MARKS PLACE  ...                 NaN                      NaN
110001   104-44 118 STREET  ...                 NaN                      NaN
110002                 NaN  ...                 NaN                      NaN
110003  8510 GLENWOOD ROAD  ...                 NaN                      NaN
110004    84-46 130 STREET  ...                 NaN                      NaN

       Road Ramp Bridge Highway Segment Garage Lot Name Ferry Direction  \
110000       NaN                    NaN             NaN             NaN
110001       NaN                    NaN             NaN             NaN
110002       NaN                    NaN             NaN             NaN
110003       NaN                    NaN             NaN             NaN
110004       NaN                    NaN             NaN             NaN

       Ferry Terminal Name   Latitude  Longitude  \
110000                 NaN  40.681541 -73.979258
110001                 NaN  40.685714 -73.824908
110002                 NaN  40.816662 -73.938991
110003                 NaN  40.639467 -73.911120
110004                 NaN  40.704888 -73.822715

                                     Location
110000   (40.68154140934097, -73.97925782435688)
110001    (40.6857138731975, -73.82490809385943)
110002    (40.816662031382485, -73.938991079402)
110003   (40.63946714255504, -73.91112033350055)
110004  (40.704888248872315, -73.82271533783924)

[5 rows x 53 columns]
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        Unique Key            Created Date             Closed Date Agency  \
120000    31488327  09/09/2015 01:59:25 AM  09/09/2015 02:45:32 AM   NYPD
120001    31491337  09/09/2015 01:52:52 AM  09/09/2015 06:17:47 AM   NYPD
120002    31487951  09/09/2015 01:51:14 AM  09/09/2015 09:17:48 AM   NYPD
120003    31487612  09/09/2015 01:50:03 AM  09/09/2015 04:56:07 AM   NYPD
120004    31488513  09/09/2015 01:48:40 AM  09/09/2015 07:46:36 AM   NYPD

                            Agency Name           Complaint Type  \
120000  New York City Police Department          Illegal Parking
120001  New York City Police Department          Illegal Parking
120002  New York City Police Department  Noise - Street/Sidewalk
120003  New York City Police Department  Noise - Street/Sidewalk
120004  New York City Police Department          Noise - Vehicle

                         Descriptor    Location Type  Incident Zip  \
120000               Blocked Hydrant  Street/Sidewalk       11228.0
120001  Commercial Overnight Parking  Street/Sidewalk       11374.0
120002                  Loud Talking  Street/Sidewalk       10456.0
120003              Loud Music/Party  Street/Sidewalk       10040.0
120004               Car/Truck Music  Street/Sidewalk       11691.0

           Incident Address  ... Bridge Highway Name Bridge Highway Direction  \
120000       1423 78 STREET  ...                 NaN                      NaN
120001       85-12 67 DRIVE  ...                 NaN                      NaN
120002  1374 COLLEGE AVENUE  ...                 NaN                      NaN
120003      93 NAGLE AVENUE  ...                 NaN                      NaN
120004   14-24 BEACON PLACE  ...                 NaN                      NaN

       Road Ramp Bridge Highway Segment Garage Lot Name Ferry Direction  \
120000       NaN                    NaN             NaN             NaN
120001       NaN                    NaN             NaN             NaN
120002       NaN                    NaN             NaN             NaN
120003       NaN                    NaN             NaN             NaN
120004       NaN                    NaN             NaN             NaN

       Ferry Terminal Name   Latitude  Longitude  \
120000                 NaN  40.616854 -74.006332
120001                 NaN  40.716282 -73.860598
120002                 NaN  40.836776 -73.911012
120003                 NaN  40.860412 -73.927936
120004                 NaN  40.611826 -73.771372

                                     Location
120000   (40.61685405880828, -74.0063322452982)
120001  (40.71628173491811, -73.86059816458915)
120002  (40.83677632719004, -73.91101214001183)
120003  (40.86041182487199, -73.92793557843983)
120004  (40.611826214529614, -73.7713717582028)

[5 rows x 53 columns]
        Unique Key            Created Date             Closed Date Agency  \
130000    31435565  08/31/2015 10:14:29 AM  08/31/2015 09:22:26 PM   NYPD
130001    31432496  08/31/2015 10:14:22 AM  08/31/2015 11:11:28 PM   NYPD
130002    31430517  08/31/2015 10:13:22 AM  08/31/2015 04:07:48 PM   NYPD
130003    31435418  08/31/2015 10:13:02 AM  08/31/2015 11:26:03 AM   NYPD
130004    31435511  08/31/2015 10:12:02 AM  08/31/2015 05:30:38 PM   NYPD

                            Agency Name           Complaint Type  \
130000  New York City Police Department  Noise - Street/Sidewalk
130001  New York City Police Department         Blocked Driveway
```

```
130002  New York City Police Department         Derelict Vehicle
130003  New York City Police Department          Illegal Parking
130004  New York City Police Department         Derelict Vehicle

                Descriptor    Location Type  Incident Zip  \
130000    Loud Music/Party  Street/Sidewalk       10032.0
130001           No Access  Street/Sidewalk       11229.0
130002  With License Plate  Street/Sidewalk       11208.0
130003    Blocked Sidewalk  Street/Sidewalk       11417.0
130004  With License Plate  Street/Sidewalk       10305.0

           Incident Address  ... Bridge Highway Name Bridge Highway Direction  \
130000  522 WEST 157 STREET  ...                 NaN                      NaN
130001        1112 AVENUE P  ...                 NaN                      NaN
130002      DANFORTH STREET  ...                 NaN                      NaN
130003     84-08 108 AVENUE  ...                 NaN                      NaN
130004    165 RADCLIFF ROAD  ...                 NaN                      NaN

       Road Ramp Bridge Highway Segment Garage Lot Name Ferry Direction  \
130000       NaN                    NaN             NaN             NaN
130001       NaN                    NaN             NaN             NaN
130002       NaN                    NaN             NaN             NaN
130003       NaN                    NaN             NaN             NaN
130004       NaN                    NaN             NaN             NaN

       Ferry Terminal Name   Latitude  Longitude  \
130000                 NaN  40.833212 -73.942882
130001                 NaN  40.610005 -73.962140
130002                 NaN  40.686977 -73.871433
130003                 NaN  40.676953 -73.852414
130004                 NaN  40.599901 -74.077797

                                     Location
130000  (40.833211919832905, -73.94288228478331)
130001   (40.61000522729272, -73.96214008211543)
130002    (40.68697747083347, -73.8714334144269)
130003   (40.67695268388007, -73.85241362532729)
130004    (40.59990074976533, -74.0777969907252)

[5 rows x 53 columns]
        Unique Key            Created Date             Closed Date Agency  \
140000    31368230  08/22/2015 11:03:17 PM  08/22/2015 11:53:27 PM   NYPD
140001    31371256  08/22/2015 11:02:36 PM  08/23/2015 05:14:02 AM   NYPD
140002    31372047  08/22/2015 11:02:19 PM  08/23/2015 03:50:07 PM   NYPD
140003    31365993  08/22/2015 11:02:19 PM  08/23/2015 03:38:01 PM   NYPD
140004    31365880  08/22/2015 11:02:12 PM  08/23/2015 02:35:48 AM   NYPD

                           Agency Name           Complaint Type  \
140000  New York City Police Department  Noise - Street/Sidewalk
140001  New York City Police Department  Noise - Street/Sidewalk
140002  New York City Police Department       Noise - Commercial
140003  New York City Police Department  Noise - Street/Sidewalk
140004  New York City Police Department         Blocked Driveway

              Descriptor     Location Type  Incident Zip     Incident Address  \
140000  Loud Music/Party   Street/Sidewalk       11203.0    76 EAST 52 STREET
140001  Loud Music/Party   Street/Sidewalk       11419.0    102-30 135 STREET
140002  Loud Music/Party  Store/Commercial       10014.0   302 WEST 14 STREET
140003  Loud Music/Party   Street/Sidewalk       10456.0  1245 FINDLAY AVENUE
140004    Partial Access   Street/Sidewalk       11374.0   64-07 BOOTH STREET
```

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        ... Bridge Highway Name Bridge Highway Direction Road Ramp  \
140000  ...                 NaN                      NaN       NaN
140001  ...                 NaN                      NaN       NaN
140002  ...                 NaN                      NaN       NaN
140003  ...                 NaN                      NaN       NaN
140004  ...                 NaN                      NaN       NaN

       Bridge Highway Segment Garage Lot Name Ferry Direction  \
140000                    NaN             NaN             NaN
140001                    NaN             NaN             NaN
140002                    NaN             NaN             NaN
140003                    NaN             NaN             NaN
140004                    NaN             NaN             NaN

       Ferry Terminal Name   Latitude  Longitude  \
140000                NaN  40.659357 -73.929252
140001                NaN  40.692498 -73.812302
140002                NaN  40.739826 -74.002717
140003                NaN  40.833736 -73.911793
140004                NaN  40.728028 -73.861504

                                     Location
140000   (40.65935745285702, -73.92925207561954)
140001   (40.69249830148893, -73.81230154321955)
140002  (40.739825531638665, -74.00271728160149)
140003  (40.833735785728855, -73.91179315290985)
140004   (40.72802766426359, -73.86150445062329)

[5 rows x 53 columns]
        Unique Key            Created Date             Closed Date Agency  \
150000    31301047  08/13/2015 10:58:41 PM  08/14/2015 12:08:12 AM   NYPD
150001    31302558  08/13/2015 10:57:45 PM  08/14/2015 04:52:24 AM   NYPD
150002    31306873  08/13/2015 10:57:41 PM  08/13/2015 11:20:22 PM   NYPD
150003    31303558  08/13/2015 10:57:36 PM  08/14/2015 07:08:52 AM   NYPD
150004    31300890  08/13/2015 10:57:33 PM  08/14/2015 02:00:33 AM   NYPD

                           Agency Name           Complaint Type  \
150000  New York City Police Department  Noise - Street/Sidewalk
150001  New York City Police Department  Noise - Street/Sidewalk
150002  New York City Police Department       Noise - Commercial
150003  New York City Police Department  Noise - Street/Sidewalk
150004  New York City Police Department          Illegal Parking

                        Descriptor        Location Type  Incident Zip  \
150000              Loud Music/Party      Street/Sidewalk       10065.0
150001              Loud Music/Party      Street/Sidewalk       10456.0
150002              Loud Music/Party  Club/Bar/Restaurant       10030.0
150003              Loud Music/Party      Street/Sidewalk       10459.0
150004  Commercial Overnight Parking      Street/Sidewalk       11220.0

            Incident Address  ... Bridge Highway Name  \
150000    330 EAST 63 STREET  ...                 NaN
150001  1296 SHERIDAN AVENUE  ...                 NaN
150002                   NaN  ...                 NaN
150003       1300 HOE AVENUE  ...                 NaN
150004         4901 9 AVENUE  ...                 NaN

       Bridge Highway Direction Road Ramp Bridge Highway Segment  \
150000                      NaN       NaN                    NaN
```

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150001                      NaN       NaN                     NaN
150002                      NaN       NaN                     NaN
150003                      NaN       NaN                     NaN
150004                      NaN       NaN                     NaN

       Garage Lot Name Ferry Direction Ferry Terminal Name   Latitude  \
150000             NaN             NaN                 NaN  40.762565
150001             NaN             NaN                 NaN  40.836252
150002             NaN             NaN                 NaN  40.817226
150003             NaN             NaN                 NaN  40.830759
150004             NaN             NaN                 NaN  40.640256

        Longitude                                 Location
150000 -73.961245  (40.76256476505412, -73.96124475967511)
150001 -73.914551  (40.83625201849241, -73.91455079032109)
150002 -73.942300  (40.81722640229732, -73.94229989188614)
150003 -73.890575  (40.83075880336612, -73.89057476850179)
150004 -74.001005  (40.64025630274153, -74.00100529922433)

[5 rows x 53 columns]
        Unique Key            Created Date             Closed Date Agency  \
160000    31238266  08/04/2015 09:17:22 PM  08/05/2015 05:47:27 AM   NYPD
160001    31236220  08/04/2015 09:16:16 PM  08/04/2015 09:40:50 PM   NYPD
160002    31239630  08/04/2015 09:12:05 PM  08/04/2015 11:25:45 PM   NYPD
160003    31236371  08/04/2015 09:12:04 PM  08/04/2015 10:54:30 PM   NYPD
160004    31236386  08/04/2015 09:11:12 PM  08/04/2015 11:59:28 PM   NYPD

                           Agency Name           Complaint Type  \
160000  New York City Police Department          Illegal Parking
160001  New York City Police Department          Illegal Parking
160002  New York City Police Department  Noise - Street/Sidewalk
160003  New York City Police Department  Noise - Street/Sidewalk
160004  New York City Police Department  Noise - Street/Sidewalk

                           Descriptor    Location Type  Incident Zip  \
160000  Double Parked Blocking Vehicle  Street/Sidewalk       10469.0
160001  Double Parked Blocking Traffic  Street/Sidewalk       10034.0
160002               Loud Music/Party  Street/Sidewalk       11216.0
160003               Loud Music/Party  Street/Sidewalk       10031.0
160004               Loud Music/Party  Street/Sidewalk       11206.0

           Incident Address  ... Bridge Highway Name Bridge Highway Direction  \
160000    1160 BURKE AVENUE  ...                 NaN                      NaN
160001   263 DYCKMAN STREET  ...                 NaN                      NaN
160002   797 STERLING PLACE  ...                 NaN                      NaN
160003  611 WEST 137 STREET  ...                 NaN                      NaN
160004   319 PULASKI STREET  ...                 NaN                      NaN

       Road Ramp Bridge Highway Segment Garage Lot Name Ferry Direction  \
160000       NaN                    NaN             NaN             NaN
160001       NaN                    NaN             NaN             NaN
160002       NaN                    NaN             NaN             NaN
160003       NaN                    NaN             NaN             NaN
160004       NaN                    NaN             NaN             NaN

       Ferry Terminal Name   Latitude  Longitude  \
160000                 NaN  40.871183 -73.853407
160001                 NaN  40.866610 -73.928551
160002                 NaN  40.672635 -73.951989
160003                 NaN  40.821612 -73.954642
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160004                 NaN  40.693653 -73.938123

                                       Location
160000   (40.87118277011055, -73.85340713651276)
160001   (40.866609736690656, -73.9285507259342)
160002   (40.67263470334773, -73.95198897690034)
160003  (40.821612489031935, -73.95464185827556)
160004   (40.693653041055796, -73.9381227756769)

[5 rows x 53 columns]
        Unique Key            Created Date             Closed Date Agency  \
170000    31166378  07/26/2015 04:11:21 PM  07/27/2015 05:04:54 AM   NYPD
170001    31166925  07/26/2015 04:09:57 PM  07/26/2015 08:39:27 PM   NYPD
170002    31168170  07/26/2015 04:09:38 PM  07/26/2015 08:41:30 PM   NYPD
170003    31168710  07/26/2015 04:09:03 PM  07/26/2015 07:52:40 PM   NYPD
170004    31165788  07/26/2015 04:07:24 PM  07/27/2015 05:28:59 AM   NYPD

                           Agency Name    Complaint Type      Descriptor  \
170000  New York City Police Department  Blocked Driveway       No Access
170001  New York City Police Department  Blocked Driveway  Partial Access
170002  New York City Police Department  Blocked Driveway       No Access
170003  New York City Police Department  Blocked Driveway       No Access
170004  New York City Police Department  Blocked Driveway       No Access

          Location Type  Incident Zip     Incident Address  ...  \
170000  Street/Sidewalk       11365.0            69 AVENUE  ...
170001  Street/Sidewalk       11421.0      86-62 85 STREET  ...
170002  Street/Sidewalk       11413.0  137-88 SLOAN STREET  ...
170003  Street/Sidewalk       11205.0     205 TAAFFE PLACE  ...
170004  Street/Sidewalk       11433.0      150-15 109 ROAD  ...

       Bridge Highway Name Bridge Highway Direction Road Ramp  \
170000                 NaN                      NaN       NaN
170001                 NaN                      NaN       NaN
170002                 NaN                      NaN       NaN
170003                 NaN                      NaN       NaN
170004                 NaN                      NaN       NaN

       Bridge Highway Segment Garage Lot Name Ferry Direction  \
170000                    NaN             NaN             NaN
170001                    NaN             NaN             NaN
170002                    NaN             NaN             NaN
170003                    NaN             NaN             NaN
170004                    NaN             NaN             NaN

       Ferry Terminal Name   Latitude  Longitude  \
170000                 NaN  40.732675 -73.793844
170001                 NaN  40.692244 -73.858997
170002                 NaN  40.674118 -73.757149
170003                 NaN  40.691986 -73.959724
170004                 NaN  40.691065 -73.797478

                                       Location
170000     (40.73267484457572, -73.793844133826)
170001  (40.692243848315826, -73.85899685690968)
170002  (40.674117625982014, -73.75714893727834)
170003    (40.69198557412709, -73.9597243777759)
170004  (40.691065099352855, -73.79747779752911)

[5 rows x 53 columns]
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        Unique Key            Created Date             Closed Date Agency  \
180000    31102593  07/17/2015 11:14:48 PM  07/18/2015 01:06:37 AM   NYPD
180001    31097902  07/17/2015 11:14:35 PM  07/18/2015 01:25:43 AM   NYPD
180002    31100921  07/17/2015 11:14:23 PM  07/18/2015 12:39:15 AM   NYPD
180003    31099540  07/17/2015 11:14:03 PM  07/17/2015 11:22:05 PM   NYPD
180004    31097848  07/17/2015 11:13:32 PM  07/18/2015 05:03:12 AM   NYPD

                            Agency Name           Complaint Type  \
180000  New York City Police Department       Noise - Commercial
180001  New York City Police Department       Noise - Commercial
180002  New York City Police Department       Noise - Commercial
180003  New York City Police Department  Noise - Street/Sidewalk
180004  New York City Police Department          Illegal Parking

                         Descriptor        Location Type  Incident Zip  \
180000              Loud Music/Party  Club/Bar/Restaurant       11238.0
180001                  Loud Talking  Club/Bar/Restaurant       10025.0
180002              Loud Music/Party  Club/Bar/Restaurant       10019.0
180003              Loud Music/Party      Street/Sidewalk       10031.0
180004  Commercial Overnight Parking      Street/Sidewalk       11385.0

            Incident Address  ... Bridge Highway Name  \
180000    1088 FULTON STREET  ...                 NaN
180001  768 AMSTERDAM AVENUE  ...                 NaN
180002         939 8 AVENUE  ...                 NaN
180003   510 WEST 136 STREET  ...                 NaN
180004       78-47 61 STREET  ...                 NaN

       Bridge Highway Direction Road Ramp Bridge Highway Segment  \
180000                      NaN       NaN                    NaN
180001                      NaN       NaN                    NaN
180002                      NaN       NaN                    NaN
180003                      NaN       NaN                    NaN
180004                      NaN       NaN                    NaN

       Garage Lot Name Ferry Direction Ferry Terminal Name   Latitude  \
180000             NaN             NaN                 NaN  40.681423
180001             NaN             NaN                 NaN  40.795136
180002             NaN             NaN                 NaN  40.765735
180003             NaN             NaN                 NaN  40.819819
180004             NaN             NaN                 NaN  40.696206

        Longitude                                 Location
180000 -73.957102  (40.681422746825206, -73.95710238584763)
180001 -73.969594   (40.79513630969814, -73.96959388942862)
180002 -73.983603  (40.765734804437116, -73.98360332757093)
180003 -73.952356    (40.8198192662946, -73.95235608572823)
180004 -73.893273   (40.69620576158944, -73.89327310042673)

[5 rows x 53 columns]
        Unique Key            Created Date             Closed Date Agency  \
190000    31037805  07/09/2015 09:54:16 AM  07/09/2015 10:42:28 AM   NYPD
190001    31039901  07/09/2015 09:54:03 AM  07/09/2015 01:11:18 PM   NYPD
190002    31038008  07/09/2015 09:53:43 AM  07/09/2015 12:48:22 PM   NYPD
190003    31039789  07/09/2015 09:53:05 AM  07/09/2015 03:36:32 PM   NYPD
190004    31039425  07/09/2015 09:51:43 AM  07/11/2015 03:48:28 PM   NYPD

                            Agency Name    Complaint Type  \
190000  New York City Police Department          Drinking
190001  New York City Police Department   Illegal Parking
```

```
190002  New York City Police Department   Noise - Vehicle
190003  New York City Police Department   Illegal Parking
190004  New York City Police Department  Blocked Driveway

                          Descriptor               Location Type  \
190000                     In Public  Residential Building/House
190001               Blocked Sidewalk             Street/Sidewalk
190002                 Car/Truck Horn             Street/Sidewalk
190003  Double Parked Blocking Vehicle             Street/Sidewalk
190004                     No Access             Street/Sidewalk

        Incident Zip     Incident Address  ... Bridge Highway Name  \
190000       11206.0  278 HUMBOLDT STREET  ...                 NaN
190001       10304.0    82 CLINTON STREET  ...                 NaN
190002       11101.0      48-15 11 STREET  ...                 NaN
190003       10462.0   2036 CRUGER AVENUE  ...                 NaN
190004       11427.0     89-97 217 STREET  ...                 NaN

       Bridge Highway Direction Road Ramp Bridge Highway Segment  \
190000                     NaN       NaN                    NaN
190001                     NaN       NaN                    NaN
190002                     NaN       NaN                    NaN
190003                     NaN       NaN                    NaN
190004                     NaN       NaN                    NaN

       Garage Lot Name Ferry Direction Ferry Terminal Name   Latitude  \
190000             NaN             NaN                 NaN  40.711235
190001             NaN             NaN                 NaN  40.632722
190002             NaN             NaN                 NaN  40.743581
190003             NaN             NaN                 NaN  40.851440
190004             NaN             NaN                 NaN  40.725377

        Longitude                                 Location
190000 -73.942125     (40.71123534973016, -73.942124719678)
190001 -74.078563  (40.632722474664305, -74.07856302179418)
190002 -73.951238   (40.74358103942738, -73.95123774166689)
190003 -73.866778  (40.851439872463665, -73.86677792790597)
190004 -73.744825   (40.72537712993866, -73.74482536875203)

[5 rows x 53 columns]
        Unique Key            Created Date             Closed Date Agency  \
200000    30969283  06/30/2015 10:24:59 AM  06/30/2015 03:39:52 PM   NYPD
200001    30969303  06/30/2015 10:24:42 AM  06/30/2015 08:08:56 PM   NYPD
200002    30971711  06/30/2015 10:23:24 AM  06/30/2015 01:45:56 PM   NYPD
200003    30977454  06/30/2015 10:23:20 AM  07/01/2015 03:17:42 PM   NYPD
200004    30976393  06/30/2015 10:23:19 AM  06/30/2015 02:38:30 PM   NYPD

                          Agency Name    Complaint Type  \
200000  New York City Police Department  Derelict Vehicle
200001  New York City Police Department   Illegal Parking
200002  New York City Police Department   Illegal Parking
200003  New York City Police Department  Derelict Vehicle
200004  New York City Police Department   Illegal Parking

                         Descriptor    Location Type  Incident Zip  \
200000             With License Plate  Street/Sidewalk       11357.0
200001  Posted Parking Sign Violation  Street/Sidewalk       10305.0
200002               Blocked Sidewalk  Street/Sidewalk       11106.0
200003             With License Plate  Street/Sidewalk       11234.0
200004  Posted Parking Sign Violation  Street/Sidewalk       11220.0
```

```
       Incident Address  ... Bridge Highway Name Bridge Highway Direction  \
200000               NaN  ...                  NaN                      NaN
200001   19 CHURCH LANE  ...                  NaN                      NaN
200002  34-40 11 STREET  ...                  NaN                      NaN
200003    5419 AVENUE K  ...                  NaN                      NaN
200004    548 52 STREET  ...                  NaN                      NaN

       Road Ramp Bridge Highway Segment Garage Lot Name Ferry Direction  \
200000       NaN                    NaN             NaN             NaN
200001       NaN                    NaN             NaN             NaN
200002       NaN                    NaN             NaN             NaN
200003       NaN                    NaN             NaN             NaN
200004       NaN                    NaN             NaN             NaN

       Ferry Terminal Name   Latitude  Longitude  \
200000                 NaN  40.784230 -73.796945
200001                 NaN  40.616706 -74.066820
200002                 NaN  40.763845 -73.938289
200003                 NaN  40.626098 -73.923480
200004                 NaN  40.643325 -74.010407

                                     Location
200000  (40.784229534491615, -73.79694526335528)
200001   (40.61670580739479, -74.06681981394985)
200002  (40.763844784756074, -73.93828859917635)
200003    (40.6260979291331, -73.92348016597818)
200004   (40.64332450636658, -74.01040658662738)

[5 rows x 53 columns]
        Unique Key            Created Date             Closed Date Agency  \
210000    30902683  06/21/2015 11:35:15 PM  06/22/2015 01:25:03 AM   NYPD
210001    30902284  06/21/2015 11:35:07 PM  06/22/2015 01:33:38 AM   NYPD
210002    30901003  06/21/2015 11:34:45 PM  06/22/2015 02:21:54 AM   NYPD
210003    30899496  06/21/2015 11:34:40 PM  06/22/2015 02:46:44 AM   NYPD
210004    30899740  06/21/2015 11:34:18 PM  06/22/2015 02:22:58 AM   NYPD

                           Agency Name           Complaint Type  \
210000  New York City Police Department          Illegal Parking
210001  New York City Police Department  Noise - Street/Sidewalk
210002  New York City Police Department  Noise - Street/Sidewalk
210003  New York City Police Department       Noise - Commercial
210004  New York City Police Department  Noise - Street/Sidewalk

                           Descriptor        Location Type  Incident Zip  \
210000  Double Parked Blocking Traffic      Street/Sidewalk       11238.0
210001                Loud Music/Party      Street/Sidewalk       11233.0
210002                Loud Music/Party      Street/Sidewalk       10040.0
210003                Loud Music/Party  Club/Bar/Restaurant       10458.0
210004                Loud Music/Party      Street/Sidewalk       10040.0

          Incident Address  ... Bridge Highway Name Bridge Highway Direction  \
210000    407 GRAND AVENUE  ...                  NaN                      NaN
210001      HANCOCK STREET  ...                  NaN                      NaN
210002   25 SHERMAN AVENUE  ...                  NaN                      NaN
210003  2334 HUGHES AVENUE  ...                  NaN                      NaN
210004  107 ELLWOOD STREET  ...                  NaN                      NaN

       Road Ramp Bridge Highway Segment Garage Lot Name Ferry Direction  \
210000       NaN                    NaN             NaN             NaN
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210001       NaN                    NaN             NaN             NaN
210002       NaN                    NaN             NaN             NaN
210003       NaN                    NaN             NaN             NaN
210004       NaN                    NaN             NaN             NaN

       Ferry Terminal Name   Latitude  Longitude  \
210000                NaN  40.683656 -73.961853
210001                NaN  40.685763 -73.921926
210002                NaN  40.862652 -73.928468
210003                NaN  40.853919 -73.887930
210004                NaN  40.861697 -73.929348

                                      Location
210000   (40.68365592000497, -73.96185308849824)
210001  (40.685762955538046, -73.92192615709062)
210002  (40.862651831098994, -73.92846821162465)
210003  (40.853918871537225, -73.88793039459689)
210004   (40.86169721771338, -73.92934773728363)

[5 rows x 53 columns]
        Unique Key            Created Date             Closed Date Agency  \
220000    30841007  06/13/2015 03:20:27 PM  06/13/2015 04:54:56 PM   NYPD
220001    30843153  06/13/2015 03:18:21 PM  06/13/2015 10:28:11 PM   NYPD
220002    30838313  06/13/2015 03:17:31 PM  06/13/2015 04:13:17 PM   NYPD
220003    30841520  06/13/2015 03:16:31 PM  06/13/2015 03:52:54 PM   NYPD
220004    30841217  06/13/2015 03:15:15 PM  06/13/2015 06:31:28 PM   NYPD

                           Agency Name           Complaint Type  \
220000  New York City Police Department  Noise - Street/Sidewalk
220001  New York City Police Department                  Traffic
220002  New York City Police Department                  Traffic
220003  New York City Police Department          Illegal Parking
220004  New York City Police Department         Blocked Driveway

                         Descriptor    Location Type  Incident Zip  \
220000               Loud Music/Party  Street/Sidewalk       11372.0
220001            Congestion/Gridlock  Street/Sidewalk       10464.0
220002            Congestion/Gridlock          Highway       10465.0
220003  Posted Parking Sign Violation  Street/Sidewalk       10306.0
220004                     No Access  Street/Sidewalk       10302.0

            Incident Address  ...    Bridge Highway Name  \
220000                   NaN  ...                    NaN
220001                   NaN  ...                    NaN
220002                   NaN  ...  Hutchinson River Pkwy
220003      2 NEUTRAL AVENUE  ...                    NaN
220004  112 TREADWELL AVENUE  ...                    NaN

       Bridge Highway Direction Road Ramp  \
220000                      NaN       NaN
220001                      NaN       NaN
220002       South/Queens Bound   Roadway
220003                      NaN       NaN
220004                      NaN       NaN

                         Bridge Highway Segment Garage Lot Name  \
220000                                       NaN             NaN
220001                                       NaN             NaN
220002  Lafayette Ave - END Hutchinson River Pkwy             NaN
220003                                       NaN             NaN
```

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220004                                        NaN             NaN

       Ferry Direction Ferry Terminal Name   Latitude  Longitude  \
220000             NaN                 NaN  40.751787 -73.884684
220001             NaN                 NaN  40.859549 -73.800902
220002             NaN                 NaN  40.816735 -73.836569
220003             NaN                 NaN  40.560995 -74.103290
220004             NaN                 NaN  40.637209 -74.138167

                                      Location
220000   (40.75178709889573, -73.88468424803324)
220001  (40.859548922129825, -73.80090160391836)
220002   (40.81673526324246, -73.83656866488622)
220003   (40.56099502203787, -74.10328983480375)
220004   (40.63720892939505, -74.13816654080291)

[5 rows x 53 columns]
        Unique Key            Created Date             Closed Date Agency  \
230000    30778464  06/05/2015 12:26:45 PM  06/05/2015 02:54:26 PM   NYPD
230001    30780268  06/05/2015 12:26:07 PM  06/05/2015 03:17:19 PM   NYPD
230002    30782418  06/05/2015 12:24:10 PM  06/05/2015 03:35:30 PM   NYPD
230003    30777977  06/05/2015 12:24:02 PM  06/05/2015 09:48:17 PM   NYPD
230004    30778017  06/05/2015 12:23:58 PM  06/05/2015 03:19:36 PM   NYPD

                           Agency Name           Complaint Type  \
230000  New York City Police Department                  Traffic
230001  New York City Police Department          Illegal Parking
230002  New York City Police Department  Noise - Street/Sidewalk
230003  New York City Police Department         Blocked Driveway
230004  New York City Police Department         Blocked Driveway

                 Descriptor    Location Type  Incident Zip  \
230000  Congestion/Gridlock  Street/Sidewalk       10310.0
230001     Blocked Sidewalk  Street/Sidewalk       11365.0
230002     Loud Music/Party  Street/Sidewalk       10031.0
230003            No Access  Street/Sidewalk       11413.0
230004            No Access  Street/Sidewalk       11223.0

          Incident Address  ... Bridge Highway Name Bridge Highway Direction  \
230000                NaN  ...                 NaN                       NaN
230001    60-14 172 STREET  ...                 NaN                       NaN
230002  295 CONVENT AVENUE  ...                 NaN                       NaN
230003   139-32 230 STREET  ...                 NaN                       NaN
230004  2030 EAST 2 STREET  ...                 NaN                       NaN

       Road Ramp Bridge Highway Segment Garage Lot Name Ferry Direction  \
230000       NaN                    NaN             NaN             NaN
230001       NaN                    NaN             NaN             NaN
230002       NaN                    NaN             NaN             NaN
230003       NaN                    NaN             NaN             NaN
230004       NaN                    NaN             NaN             NaN

       Ferry Terminal Name   Latitude  Longitude  \
230000                NaN  40.624379 -74.124746
230001                NaN  40.738623 -73.797286
230002                NaN  40.822864 -73.947516
230003                NaN  40.668794 -73.744988
230004                NaN  40.600203 -73.970172

                                      Location
```

```
230000  (40.624378647461945, -74.12474636945765)
230001  (40.738623362244155, -73.79728633027798)
230002   (40.82286380172492, -73.94751594978828)
230003  (40.668793873314755, -73.74498777202152)
230004  (40.600203180526236, -73.97017240692787)

[5 rows x 53 columns]
        Unique Key            Created Date             Closed Date Agency  \
240000    30711912  05/27/2015 04:18:00 PM  05/27/2015 06:29:55 PM   NYPD
240001    30712199  05/27/2015 04:17:29 PM  05/27/2015 11:31:04 PM   NYPD
240002    30706745  05/27/2015 04:17:09 PM  05/27/2015 08:00:40 PM   NYPD
240003    30711014  05/27/2015 04:12:27 PM  05/27/2015 04:26:06 PM   NYPD
240004    30711933  05/27/2015 04:12:01 PM  05/27/2015 04:31:38 PM   NYPD

                            Agency Name           Complaint Type  \
240000  New York City Police Department      Homeless Encampment
240001  New York City Police Department          Noise - Vehicle
240002  New York City Police Department         Derelict Vehicle
240003  New York City Police Department                  Traffic
240004  New York City Police Department  Noise - Street/Sidewalk

                Descriptor               Location Type  Incident Zip  \
240000                 NaN  Residential Building/House       10003.0
240001     Car/Truck Music             Street/Sidewalk       11226.0
240002   With License Plate             Street/Sidewalk       11378.0
240003  Congestion/Gridlock             Street/Sidewalk       11216.0
240004     Loud Music/Party             Street/Sidewalk       11211.0

           Incident Address  ... Bridge Highway Name Bridge Highway Direction  \
240000    32 EAST 13 STREET  ...                 NaN                      NaN
240001  184 CLARKSON AVENUE  ...                 NaN                      NaN
240002                 NaN  ...                 NaN                      NaN
240003      ATLANTIC AVENUE  ...                 NaN                      NaN
240004                 NaN  ...                 NaN                      NaN

       Road Ramp Bridge Highway Segment Garage Lot Name Ferry Direction  \
240000       NaN                    NaN             NaN             NaN
240001       NaN                    NaN             NaN             NaN
240002       NaN                    NaN             NaN             NaN
240003       NaN                    NaN             NaN             NaN
240004       NaN                    NaN             NaN             NaN

       Ferry Terminal Name   Latitude  Longitude  \
240000                 NaN  40.734734 -73.992765
240001                 NaN  40.655181 -73.953977
240002                 NaN  40.724638 -73.887076
240003                 NaN  40.678237 -73.945535
240004                 NaN  40.710264 -73.953856

                                     Location
240000  (40.73473382032559, -73.99276529442523)
240001   (40.65518146681626, -73.9539765456767)
240002  (40.72463835706342, -73.88707628754645)
240003  (40.67823666015777, -73.94553510571777)
240004  (40.71026351414386, -73.95385555332841)

[5 rows x 53 columns]
        Unique Key            Created Date             Closed Date Agency  \
250000    30648774  05/18/2015 06:07:04 PM  05/18/2015 06:59:42 PM   NYPD
250001    30644637  05/18/2015 06:06:35 PM  05/18/2015 08:30:42 PM   NYPD
```

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250002    30648764  05/18/2015 06:02:44 PM  05/18/2015 06:10:26 PM   NYPD
250003    30643704  05/18/2015 06:01:35 PM  05/18/2015 08:30:41 PM   NYPD
250004    30646024  05/18/2015 06:00:43 PM  05/19/2015 02:03:49 AM   NYPD

                           Agency Name      Complaint Type        Descriptor  \
250000  New York City Police Department  Noise - Commercial  Loud Music/Party
250001  New York City Police Department     Noise - Vehicle   Car/Truck Music
250002  New York City Police Department  Noise - Commercial  Loud Music/Party
250003  New York City Police Department    Blocked Driveway         No Access
250004  New York City Police Department    Blocked Driveway         No Access

              Location Type  Incident Zip            Incident Address  ...  \
250000     Store/Commercial       10032.0               3810 BROADWAY  ...
250001      Street/Sidewalk       10458.0  358 BEDFORD PARK BOULEVARD  ...
250002  Club/Bar/Restaurant       10019.0                737 9 AVENUE  ...
250003      Street/Sidewalk       10467.0            3273 HULL AVENUE  ...
250004      Street/Sidewalk       11378.0             55-62 64 STREET  ...

       Bridge Highway Name Bridge Highway Direction Road Ramp  \
250000                NaN                      NaN       NaN
250001                NaN                      NaN       NaN
250002                NaN                      NaN       NaN
250003                NaN                      NaN       NaN
250004                NaN                      NaN       NaN

       Bridge Highway Segment Garage Lot Name Ferry Direction  \
250000                    NaN             NaN             NaN
250001                    NaN             NaN             NaN
250002                    NaN             NaN             NaN
250003                    NaN             NaN             NaN
250004                    NaN             NaN             NaN

       Ferry Terminal Name   Latitude  Longitude  \
250000                NaN  40.835016 -73.944167
250001                NaN  40.868386 -73.884558
250002                NaN  40.763348 -73.989058
250003                NaN  40.876080 -73.874988
250004                NaN  40.725294 -73.901799

                                     Location
250000   (40.83501582494275, -73.9441672387562)
250001  (40.86838561054362, -73.88455797980147)
250002  (40.76334752620899, -73.98905844724008)
250003   (40.8760800009819, -73.87498751672649)
250004  (40.72529412843385, -73.90179870392225)

[5 rows x 53 columns]
        Unique Key            Created Date             Closed Date Agency  \
260000    30589935  05/10/2015 03:14:44 AM  05/10/2015 04:48:46 AM   NYPD
260001    30588333  05/10/2015 03:14:26 AM  05/10/2015 07:49:45 AM   NYPD
260002    30590469  05/10/2015 03:12:53 AM  05/10/2015 03:57:36 AM   NYPD
260003    30587041  05/10/2015 03:12:46 AM  05/10/2015 05:03:58 AM   NYPD
260004    30589274  05/10/2015 03:11:59 AM  05/10/2015 07:01:17 AM   NYPD

                           Agency Name           Complaint Type  \
260000  New York City Police Department  Noise - Street/Sidewalk
260001  New York City Police Department       Noise - Commercial
260002  New York City Police Department          Noise - Vehicle
260003  New York City Police Department             Noise - Park
260004  New York City Police Department          Noise - Vehicle
```

```
              Descriptor        Location Type  Incident Zip  \
260000      Loud Talking      Street/Sidewalk       10035.0
260001  Loud Music/Party  Club/Bar/Restaurant       11215.0
260002   Car/Truck Music      Street/Sidewalk       11212.0
260003      Loud Talking      Park/Playground       11204.0
260004   Car/Truck Music      Street/Sidewalk       11220.0

           Incident Address  ... Bridge Highway Name Bridge Highway Direction  \
260000  306 EAST 126 STREET  ...                 NaN                      NaN
260001                  NaN  ...                 NaN                      NaN
260002                  NaN  ...                 NaN                      NaN
260003                  NaN  ...                 NaN                      NaN
260004        418 61 STREET  ...                 NaN                      NaN

       Road Ramp Bridge Highway Segment Garage Lot Name Ferry Direction  \
260000       NaN                    NaN             NaN             NaN
260001       NaN                    NaN             NaN             NaN
260002       NaN                    NaN             NaN             NaN
260003       NaN                    NaN             NaN             NaN
260004       NaN                    NaN             NaN             NaN

       Ferry Terminal Name   Latitude  Longitude  \
260000                 NaN  40.803182 -73.932531
260001                 NaN  40.667254 -73.994543
260002                 NaN  40.663294 -73.925424
260003                 NaN  40.624322 -73.984964
260004                 NaN  40.639846 -74.018347

                                     Location
260000   (40.80318184014734, -73.93253065294712)
260001     (40.66725385045012, -73.994542512279)
260002     (40.6632937941437, -73.9254235594002)
260003    (40.62432182804097, -73.9849637376358)
260004  (40.639845870326205, -74.01834749893119)

[5 rows x 53 columns]
        Unique Key            Created Date             Closed Date Agency  \
270000    30525903  05/01/2015 10:22:56 PM  05/01/2015 10:40:55 PM   NYPD
270001    30526608  05/01/2015 10:22:41 PM  05/02/2015 12:36:18 PM   NYPD
270002    30527584  05/01/2015 10:21:54 PM  05/01/2015 11:06:26 PM   NYPD
270003    30525680  05/01/2015 10:21:44 PM  05/02/2015 12:19:48 AM   NYPD
270004    30525902  05/01/2015 10:21:39 PM  05/02/2015 02:11:11 AM   NYPD

                          Agency Name      Complaint Type        Descriptor  \
270000  New York City Police Department  Noise - Commercial  Loud Music/Party
270001  New York City Police Department    Blocked Driveway         No Access
270002  New York City Police Department  Noise - Commercial      Loud Talking
270003  New York City Police Department  Noise - Commercial  Loud Music/Party
270004  New York City Police Department  Noise - Commercial  Loud Music/Party

              Location Type  Incident Zip      Incident Address  ...  \
270000  Club/Bar/Restaurant       10011.0          174 8 AVENUE  ...
270001      Street/Sidewalk       10469.0    1058 DUNCAN STREET  ...
270002     Store/Commercial       11427.0      89-91 217 STREET  ...
270003  Club/Bar/Restaurant       10012.0  333 LAFAYETTE STREET  ...
270004  Club/Bar/Restaurant       10303.0   163 VAN PELT AVENUE  ...

       Bridge Highway Name Bridge Highway Direction Road Ramp  \
270000                 NaN                      NaN       NaN
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270001                NaN                      NaN       NaN
270002                NaN                      NaN       NaN
270003                NaN                      NaN       NaN
270004                NaN                      NaN       NaN

       Bridge Highway Segment Garage Lot Name Ferry Direction  \
270000                    NaN             NaN             NaN
270001                    NaN             NaN             NaN
270002                    NaN             NaN             NaN
270003                    NaN             NaN             NaN
270004                    NaN             NaN             NaN

       Ferry Terminal Name   Latitude  Longitude  \
270000                NaN  40.742688 -74.000372
270001                NaN  40.873032 -73.857055
270002                NaN  40.725473 -73.744926
270003                NaN  40.725761 -73.994776
270004                NaN  40.632400 -74.155176

                                     Location
270000   (40.74268833848282, -74.00037170256458)
270001    (40.87303178295427, -73.8570550602614)
270002   (40.72547341986226, -73.74492601973732)
270003     (40.725761324016, -73.99477583879312)
270004  (40.632400315977364, -74.15517623099356)

[5 rows x 53 columns]
        Unique Key            Created Date             Closed Date Agency  \
280000    30450906  04/21/2015 10:50:46 AM  04/21/2015 04:04:07 PM   NYPD
280001    30452724  04/21/2015 10:49:37 AM  04/21/2015 01:32:21 PM   NYPD
280002    30452354  04/21/2015 10:45:29 AM  04/21/2015 03:20:47 PM   NYPD
280003    30448866  04/21/2015 10:40:51 AM  04/21/2015 03:05:35 PM   NYPD
280004    30446786  04/21/2015 10:38:48 AM  04/21/2015 11:12:59 AM   NYPD

                           Agency Name      Complaint Type  \
280000  New York City Police Department     Illegal Parking
280001  New York City Police Department  Noise - Commercial
280002  New York City Police Department    Blocked Driveway
280003  New York City Police Department     Illegal Parking
280004  New York City Police Department     Illegal Parking

                           Descriptor     Location Type  Incident Zip  \
280000  Double Parked Blocking Vehicle   Street/Sidewalk       10034.0
280001                Loud Music/Party  Store/Commercial       10018.0
280002                       No Access   Street/Sidewalk       11420.0
280003                 Blocked Hydrant   Street/Sidewalk       11432.0
280004  Double Parked Blocking Traffic   Street/Sidewalk       11235.0

         Incident Address  ... Bridge Highway Name Bridge Highway Direction  \
280000  25 CUMMING STREET  ...                 NaN                      NaN
280001      1385 BROADWAY  ...                 NaN                      NaN
280002  115-23 121 STREET  ...                 NaN                      NaN
280003   82-75 160 STREET  ...                 NaN                      NaN
280004   54 PARKWAY COURT  ...                 NaN                      NaN

       Road Ramp Bridge Highway Segment Garage Lot Name Ferry Direction  \
280000       NaN                    NaN             NaN             NaN
280001       NaN                    NaN             NaN             NaN
280002       NaN                    NaN             NaN             NaN
280003       NaN                    NaN             NaN             NaN
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280004       NaN                    NaN             NaN             NaN

       Ferry Terminal Name   Latitude  Longitude  \
280000                NaN  40.866699 -73.926486
280001                NaN  40.752613 -73.987389
280002                NaN  40.676927 -73.817569
280003                NaN  40.716683 -73.805845
280004                NaN  40.589725 -73.964629

                                     Location
280000    (40.8666990148464, -73.92648618310697)
280001  (40.752612659697014, -73.98738910435912)
280002    (40.67692741509003, -73.8175693195189)
280003   (40.71668274973942, -73.80584479598353)
280004  (40.589724710447705, -73.96462872184478)

[5 rows x 53 columns]
        Unique Key            Created Date             Closed Date Agency  \
290000    30382337  04/11/2015 09:33:31 AM  04/11/2015 10:03:29 AM   NYPD
290001    30381499  04/11/2015 09:32:31 AM  04/11/2015 10:37:02 AM   NYPD
290002    30377636  04/11/2015 09:31:01 AM  04/11/2015 04:28:11 PM   NYPD
290003    30378484  04/11/2015 09:30:33 AM  04/11/2015 01:59:50 PM   NYPD
290004    30377503  04/11/2015 09:29:03 AM  04/11/2015 12:46:09 PM   NYPD

                           Agency Name      Complaint Type  \
290000  New York City Police Department     Illegal Parking
290001  New York City Police Department  Noise - Commercial
290002  New York City Police Department    Derelict Vehicle
290003  New York City Police Department    Blocked Driveway
290004  New York City Police Department             Traffic

                        Descriptor     Location Type  Incident Zip  \
290000  Commercial Overnight Parking   Street/Sidewalk       11230.0
290001                 Loud Talking  Store/Commercial       10002.0
290002            With License Plate   Street/Sidewalk       11429.0
290003                     No Access   Street/Sidewalk       11229.0
290004           Congestion/Gridlock   Street/Sidewalk       10128.0

           Incident Address  ... Bridge Highway Name Bridge Highway Direction  \
290000     5009 BAY PARKWAY  ...                 NaN                      NaN
290001    211 EAST BROADWAY  ...                 NaN                      NaN
290002     99-30 211 STREET  ...                 NaN                      NaN
290003  2016 EAST 15 STREET  ...                 NaN                      NaN
290004       EAST 89 STREET  ...                 NaN                      NaN

       Road Ramp Bridge Highway Segment Garage Lot Name Ferry Direction  \
290000       NaN                    NaN             NaN             NaN
290001       NaN                    NaN             NaN             NaN
290002       NaN                    NaN             NaN             NaN
290003       NaN                    NaN             NaN             NaN
290004       NaN                    NaN             NaN             NaN

       Ferry Terminal Name   Latitude  Longitude  \
290000                NaN  40.621778 -73.974503
290001                NaN  40.714263 -73.987667
290002                NaN  40.713045 -73.748455
290003                NaN  40.600490 -73.956729
290004                NaN  40.781745 -73.954933

                                     Location
```

```
290000  (40.621778307544346, -73.97450344744206)
290001  (40.714262985992505, -73.98766690196925)
290002   (40.71304462643682, -73.74845450431472)
290003   (40.60048985383589, -73.95672945733943)
290004   (40.78174545031244, -73.95493261021181)

[5 rows x 53 columns]
        Unique Key            Created Date             Closed Date Agency  \
300000    30286190  03/29/2015 10:01:36 PM  03/30/2015 12:12:45 AM   NYPD
300001    30284907  03/29/2015 09:58:52 PM  03/29/2015 11:11:01 PM   NYPD
300002    30288690  03/29/2015 09:57:32 PM                     NaN   NYPD
300003    30287402  03/29/2015 09:55:05 PM  03/29/2015 10:46:20 PM   NYPD
300004    30288493  03/29/2015 09:52:31 PM  03/29/2015 10:58:10 PM   NYPD

                           Agency Name           Complaint Type  \
300000  New York City Police Department       Noise - Commercial
300001  New York City Police Department          Illegal Parking
300002  New York City Police Department          Illegal Parking
300003  New York City Police Department  Noise - Street/Sidewalk
300004  New York City Police Department       Noise - Commercial

              Descriptor        Location Type  Incident Zip  \
300000  Loud Music/Party  Club/Bar/Restaurant       10002.0
300001   Blocked Hydrant      Street/Sidewalk       11419.0
300002   Blocked Hydrant      Street/Sidewalk           NaN
300003      Loud Talking      Street/Sidewalk       11211.0
300004  Loud Music/Party  Club/Bar/Restaurant       11374.0

                 Incident Address  ... Bridge Highway Name  \
300000         75A ORCHARD STREET  ...                 NaN
300001                        NaN  ...                 NaN
300002    67-52-67-7 BOOTH STREET  ...                 NaN
300003         245 SOUTH 3 STREET  ...                 NaN
300004  67-63 WOODHAVEN BOULEVARD  ...                 NaN

       Bridge Highway Direction Road Ramp Bridge Highway Segment  \
300000                      NaN       NaN                    NaN
300001                      NaN       NaN                    NaN
300002                      NaN       NaN                    NaN
300003                      NaN       NaN                    NaN
300004                      NaN       NaN                    NaN

       Garage Lot Name Ferry Direction Ferry Terminal Name   Latitude  \
300000             NaN             NaN                 NaN  40.717576
300001             NaN             NaN                 NaN  40.689476
300002             NaN             NaN                 NaN        NaN
300003             NaN             NaN                 NaN  40.710737
300004             NaN             NaN                 NaN  40.715865

        Longitude                                 Location
300000 -73.990523  (40.71757618265162, -73.99052335213463)
300001 -73.827686  (40.68947566409738, -73.82768556855798)
300002        NaN                                       NaN
300003 -73.957625  (40.71073706249216, -73.95762457540624)
300004 -73.861021  (40.71586504212003, -73.86102109195703)

[5 rows x 53 columns]
        Unique Key            Created Date             Closed Date Agency  \
310000    30199835  03/17/2015 10:12:42 AM  03/17/2015 10:30:11 AM   NYPD
310001    30194041  03/17/2015 10:11:55 AM  03/17/2015 10:29:57 AM   NYPD
```

```
310002    30201205  03/17/2015 10:10:54 AM  03/17/2015 11:43:38 AM   NYPD
310003    30199250  03/17/2015 10:10:02 AM  03/18/2015 01:03:28 AM   NYPD
310004    30195838  03/17/2015 10:09:16 AM  03/17/2015 02:17:23 PM   NYPD

                           Agency Name       Complaint Type  \
310000  New York City Police Department     Derelict Vehicle
310001  New York City Police Department     Derelict Vehicle
310002  New York City Police Department  Homeless Encampment
310003  New York City Police Department     Blocked Driveway
310004  New York City Police Department     Blocked Driveway

                Descriptor    Location Type  Incident Zip  \
310000  With License Plate  Street/Sidewalk       11229.0
310001  With License Plate  Street/Sidewalk       11223.0
310002                 NaN  Street/Sidewalk       11222.0
310003           No Access  Street/Sidewalk       10472.0
310004      Partial Access  Street/Sidewalk       11205.0

             Incident Address  ... Bridge Highway Name  \
310000          2934 AVENUE T  ...                 NaN
310001           851 AVENUE W  ...                 NaN
310002                    NaN  ...                 NaN
310003  1143 UNDERHILL AVENUE  ...                 NaN
310004        777 KENT AVENUE  ...                 NaN

       Bridge Highway Direction Road Ramp Bridge Highway Segment  \
310000                      NaN       NaN                    NaN
310001                      NaN       NaN                    NaN
310002                      NaN       NaN                    NaN
310003                      NaN       NaN                    NaN
310004                      NaN       NaN                    NaN

       Garage Lot Name Ferry Direction Ferry Terminal Name   Latitude  \
310000             NaN             NaN                 NaN  40.602707
310001             NaN             NaN                 NaN  40.593665
310002             NaN             NaN                 NaN  40.721380
310003             NaN             NaN                 NaN  40.829254
310004             NaN             NaN                 NaN  40.697574

        Longitude                                 Location
310000 -73.942042   (40.60270667146611, -73.94204233867774)
310001 -73.961616  (40.593665276272965, -73.96161644155536)
310002 -73.947949   (40.72138002248175, -73.94794941491676)
310003 -73.861951   (40.82925417936658, -73.86195140172586)
310004 -73.959999  (40.697574018345165, -73.95999868851226)

[5 rows x 53 columns]
        Unique Key            Created Date             Closed Date Agency  \
320000    30107458  03/05/2015 05:31:28 PM  03/06/2015 12:02:20 AM   NYPD
320001    30104405  03/05/2015 05:28:12 PM  03/06/2015 12:10:44 AM   NYPD
320002    30106011  03/05/2015 05:25:05 PM  03/05/2015 08:46:28 PM   NYPD
320003    30104970  03/05/2015 05:24:31 PM  03/05/2015 06:12:51 PM   NYPD
320004    30106974  03/05/2015 05:23:53 PM  03/05/2015 07:34:27 PM   NYPD

                           Agency Name    Complaint Type  \
320000  New York City Police Department  Blocked Driveway
320001  New York City Police Department  Blocked Driveway
320002  New York City Police Department   Illegal Parking
320003  New York City Police Department      Animal Abuse
320004  New York City Police Department   Illegal Parking
```

```
                            Descriptor               Location Type  \
320000                       No Access             Street/Sidewalk
320001                       No Access             Street/Sidewalk
320002  Double Parked Blocking Traffic             Street/Sidewalk
320003      Other (complaint details)  Residential Building/House
320004   Posted Parking Sign Violation             Street/Sidewalk

        Incident Zip      Incident Address  ... Bridge Highway Name  \
320000       10453.0  1722 TOWNSEND AVENUE  ...                 NaN
320001       11219.0        1528 57 STREET  ...                 NaN
320002       11234.0                   NaN  ...                 NaN
320003       10019.0    544 WEST 49 STREET  ...                 NaN
320004       10018.0    312 WEST 37 STREET  ...                 NaN

       Bridge Highway Direction Road Ramp Bridge Highway Segment  \
320000                      NaN       NaN                    NaN
320001                      NaN       NaN                    NaN
320002                      NaN       NaN                    NaN
320003                      NaN       NaN                    NaN
320004                      NaN       NaN                    NaN

       Garage Lot Name Ferry Direction Ferry Terminal Name   Latitude  \
320000             NaN             NaN                 NaN  40.846729
320001             NaN             NaN                 NaN  40.627413
320002             NaN             NaN                 NaN  40.620152
320003             NaN             NaN                 NaN  40.764750
320004             NaN             NaN                 NaN  40.754367

        Longitude                                 Location
320000 -73.911881  (40.846729310440374, -73.91188074098326)
320001 -73.991613   (40.627413139142924, -73.9916133109718)
320002 -73.907796   (40.62015217914993, -73.90779606279581)
320003 -73.993917   (40.76475044839946, -73.99391722048276)
320004 -73.992752   (40.75436701489852, -73.99275233391495)

[5 rows x 53 columns]
        Unique Key            Created Date             Closed Date Agency  \
330000    29995827  02/21/2015 01:19:32 AM  02/21/2015 05:05:42 AM   NYPD
330001    30002190  02/21/2015 01:16:28 AM  02/21/2015 07:03:32 AM   NYPD
330002    29995587  02/21/2015 01:16:18 AM  02/21/2015 06:07:06 AM   NYPD
330003    29998530  02/21/2015 01:14:25 AM  02/21/2015 05:18:05 AM   NYPD
330004    29996753  02/21/2015 01:13:52 AM  02/21/2015 07:34:35 AM   NYPD

                            Agency Name      Complaint Type  \
330000  New York City Police Department        Animal Abuse
330001  New York City Police Department  Noise - Commercial
330002  New York City Police Department    Derelict Vehicle
330003  New York City Police Department    Blocked Driveway
330004  New York City Police Department     Noise - Vehicle

                Descriptor        Location Type  Incident Zip  \
330000          No Shelter      House and Store       11368.0
330001    Banging/Pounding  Club/Bar/Restaurant       11216.0
330002  With License Plate      Street/Sidewalk       11422.0
330003           No Access      Street/Sidewalk       11207.0
330004      Car/Truck Horn      Street/Sidewalk       10475.0

             Incident Address  ... Bridge Highway Name  \
330000       34-26 106 STREET  ...                 NaN
```

```
330001    761 NOSTRAND AVENUE  ...                 NaN
330002  155-60 BAYVIEW AVENUE  ...                 NaN
330003    737 CHAUNCEY STREET  ...                 NaN
330004     3300 PALMER AVENUE  ...                 NaN

       Bridge Highway Direction Road Ramp Bridge Highway Segment  \
330000                     NaN       NaN                    NaN
330001                     NaN       NaN                    NaN
330002                     NaN       NaN                    NaN
330003                     NaN       NaN                    NaN
330004                     NaN       NaN                    NaN

       Garage Lot Name Ferry Direction Ferry Terminal Name   Latitude  \
330000             NaN             NaN                 NaN  40.755328
330001             NaN             NaN                 NaN  40.671278
330002             NaN             NaN                 NaN  40.637471
330003             NaN             NaN                 NaN  40.685343
330004             NaN             NaN                 NaN  40.878305

        Longitude                                 Location
330000 -73.862375   (40.75532803460063, -73.86237531851633)
330001 -73.950357   (40.67127809120883, -73.95035693397014)
330002 -73.745410   (40.63747122563547, -73.74541013177596)
330003 -73.907410  (40.685343220141434, -73.90741025722555)
330004 -73.834946  (40.878305031663054, -73.83494555762553)

[5 rows x 53 columns]
        Unique Key            Created Date             Closed Date Agency  \
340000    29884274  02/06/2015 11:56:56 PM  02/07/2015 01:40:03 AM   NYPD
340001    29888639  02/06/2015 11:56:12 PM  02/07/2015 04:14:18 AM   NYPD
340002    29883579  02/06/2015 11:53:55 PM  02/07/2015 07:33:15 AM   NYPD
340003    29882163  02/06/2015 11:53:06 PM  02/07/2015 01:18:48 AM   NYPD
340004    29882138  02/06/2015 11:52:43 PM  02/07/2015 01:16:42 AM   NYPD

                           Agency Name           Complaint Type  \
340000  New York City Police Department       Noise - Commercial
340001  New York City Police Department       Noise - Commercial
340002  New York City Police Department         Blocked Driveway
340003  New York City Police Department  Noise - Street/Sidewalk
340004  New York City Police Department       Noise - Commercial

              Descriptor        Location Type  Incident Zip  \
340000  Loud Music/Party  Club/Bar/Restaurant       11217.0
340001  Loud Music/Party     Store/Commercial       10002.0
340002         No Access      Street/Sidewalk       11436.0
340003      Loud Talking      Street/Sidewalk       10032.0
340004   Car/Truck Music     Store/Commercial       11377.0

           Incident Address  ... Bridge Highway Name Bridge Highway Direction  \
340000  80 LAFAYETTE AVENUE  ...                 NaN                      NaN
340001   102 NORFOLK STREET  ...                 NaN                      NaN
340002    128-33 140 STREET  ...                 NaN                      NaN
340003  580 WEST 172 STREET  ...                 NaN                      NaN
340004      38-12 65 STREET  ...                 NaN                      NaN

       Road Ramp Bridge Highway Segment Garage Lot Name Ferry Direction  \
340000       NaN                    NaN             NaN             NaN
340001       NaN                    NaN             NaN             NaN
340002       NaN                    NaN             NaN             NaN
340003       NaN                    NaN             NaN             NaN
```

```
340004       NaN                    NaN             NaN             NaN

       Ferry Terminal Name   Latitude  Longitude  \
340000                NaN  40.687167 -73.974760
340001                NaN  40.718819 -73.987125
340002                NaN  40.672072 -73.799684
340003                NaN  40.843268 -73.937452
340004                NaN  40.746775 -73.898703

                                      Location
340000   (40.68716729285692, -73.97475969401174)
340001   (40.71881922895107, -73.98712493829669)
340002  (40.672072339470795, -73.79968402029776)
340003   (40.84326846632358, -73.93745232028013)
340004  (40.746774643319874, -73.89870301132463)

[5 rows x 53 columns]
        Unique Key            Created Date             Closed Date Agency  \
350000    29777838  01/23/2015 03:54:05 PM  01/24/2015 09:13:38 AM   NYPD
350001    29780154  01/23/2015 03:53:18 PM  01/23/2015 06:02:56 PM   NYPD
350002    29777258  01/23/2015 03:53:10 PM  01/23/2015 08:53:54 PM   NYPD
350003    29781204  01/23/2015 03:50:33 PM  01/24/2015 09:13:40 AM   NYPD
350004    29775445  01/23/2015 03:48:21 PM  01/23/2015 04:59:43 PM   NYPD

                           Agency Name    Complaint Type      Descriptor  \
350000  New York City Police Department  Blocked Driveway       No Access
350001  New York City Police Department  Blocked Driveway       No Access
350002  New York City Police Department  Blocked Driveway  Partial Access
350003  New York City Police Department  Blocked Driveway       No Access
350004  New York City Police Department  Blocked Driveway       No Access

          Location Type  Incident Zip   Incident Address  ...  \
350000  Street/Sidewalk       11415.0   84-28 130 STREET  ...
350001  Street/Sidewalk       11228.0     1218 76 STREET  ...
350002  Street/Sidewalk       11214.0  180 BAY 26 STREET  ...
350003  Street/Sidewalk       11415.0   84-22 130 STREET  ...
350004  Street/Sidewalk       11373.0    41-25 77 STREET  ...

       Bridge Highway Name Bridge Highway Direction Road Ramp  \
350000                NaN                      NaN       NaN
350001                NaN                      NaN       NaN
350002                NaN                      NaN       NaN
350003                NaN                      NaN       NaN
350004                NaN                      NaN       NaN

       Bridge Highway Segment Garage Lot Name Ferry Direction  \
350000                    NaN             NaN             NaN
350001                    NaN             NaN             NaN
350002                    NaN             NaN             NaN
350003                    NaN             NaN             NaN
350004                    NaN             NaN             NaN

       Ferry Terminal Name   Latitude  Longitude  \
350000                NaN  40.705317 -73.822801
350001                NaN  40.620655 -74.009650
350002                NaN  40.600031 -74.000648
350003                NaN  40.705432 -73.822822
350004                NaN  40.744783 -73.888211

                                      Location
```

```
350000    (40.7053165645494, -73.8228007606856)
350001  (40.62065536535492, -74.00965019982057)
350002   (40.6000313824242, -74.00064818940417)
350003  (40.70543187773143, -73.82282209468325)
350004  (40.74478322774275, -73.88821141933026)

[5 rows x 53 columns]
        Unique Key            Created Date             Closed Date Agency  \
360000    29661353  01/07/2015 09:24:22 PM  01/07/2015 09:46:29 PM   NYPD
360001    29655462  01/07/2015 09:17:58 PM  01/08/2015 01:28:18 AM   NYPD
360002    29658905  01/07/2015 09:17:23 PM  01/08/2015 08:17:34 AM   NYPD
360003    29653723  01/07/2015 09:15:18 PM  01/07/2015 09:23:25 PM   NYPD
360004    29656496  01/07/2015 09:14:37 PM  01/07/2015 10:14:38 PM   NYPD

                           Agency Name    Complaint Type  \
360000  New York City Police Department   Illegal Parking
360001  New York City Police Department  Blocked Driveway
360002  New York City Police Department      Animal Abuse
360003  New York City Police Department   Illegal Parking
360004  New York City Police Department   Noise - Vehicle

                           Descriptor               Location Type  \
360000    Commercial Overnight Parking             Street/Sidewalk
360001                 Partial Access             Street/Sidewalk
360002                      Neglected  Residential Building/House
360003  Double Parked Blocking Traffic             Street/Sidewalk
360004                 Car/Truck Horn             Street/Sidewalk

        Incident Zip    Incident Address  ... Bridge Highway Name  \
360000       11223.0   390 KINGS HIGHWAY  ...                 NaN
360001       11203.0  3316 SNYDER AVENUE  ...                 NaN
360002       10457.0  1751 MONROE AVENUE  ...                 NaN
360003       10014.0   538 HUDSON STREET  ...                 NaN
360004       10013.0                 NaN  ...                 NaN

       Bridge Highway Direction Road Ramp Bridge Highway Segment  \
360000                      NaN       NaN                    NaN
360001                      NaN       NaN                    NaN
360002                      NaN       NaN                    NaN
360003                      NaN       NaN                    NaN
360004                      NaN       NaN                    NaN

       Garage Lot Name Ferry Direction Ferry Terminal Name   Latitude  \
360000             NaN             NaN                 NaN  40.604580
360001             NaN             NaN                 NaN  40.649109
360002             NaN             NaN                 NaN  40.845937
360003             NaN             NaN                 NaN  40.734544
360004             NaN             NaN                 NaN  40.719580

        Longitude                                 Location
360000 -73.975731   (40.60457969491128, -73.97573085537698)
360001 -73.946222  (40.649109387875534, -73.94622196443328)
360002 -73.906561   (40.84593739332507, -73.90656144332074)
360003 -74.006152  (40.734544495244954, -74.00615218850524)
360004 -73.998056  (40.719580230801505, -73.99805555128263)

[5 rows x 53 columns]
```

Exploring the Data

In [3]:
```
df = pd.read_csv("311_Service_Requests_from_2010_to_Present.csv", low_memory=Fal

df.head()
```

Out[3]:

	Unique Key	Created Date	Closed Date	Agency	Agency Name	Complaint Type	Descriptor
0	32310363	12/31/2015 11:59:45 PM	01/01/2016 12:55:15 AM	NYPD	New York City Police Department	Noise - Street/Sidewalk	Loud Music/Party
1	32309934	12/31/2015 11:59:44 PM	01/01/2016 01:26:57 AM	NYPD	New York City Police Department	Blocked Driveway	No Access
2	32309159	12/31/2015 11:59:29 PM	01/01/2016 04:51:03 AM	NYPD	New York City Police Department	Blocked Driveway	No Access
3	32305098	12/31/2015 11:57:46 PM	01/01/2016 07:43:13 AM	NYPD	New York City Police Department	Illegal Parking	Commercial Overnight Parking
4	32306529	12/31/2015 11:56:58 PM	01/01/2016 03:24:42 AM	NYPD	New York City Police Department	Illegal Parking	Blocked Sidewalk

5 rows × 53 columns

In [4]:
```
print(df.shape)
```

```
(364558, 53)
```

In [5]:
```
print("Columns:\n", df.columns)
```

```
Columns:
 Index(['Unique Key', 'Created Date', 'Closed Date', 'Agency', 'Agency Name',
       'Complaint Type', 'Descriptor', 'Location Type', 'Incident Zip',
       'Incident Address', 'Street Name', 'Cross Street 1', 'Cross Street 2',
       'Intersection Street 1', 'Intersection Street 2', 'Address Type',
       'City', 'Landmark', 'Facility Type', 'Status', 'Due Date',
       'Resolution Description', 'Resolution Action Updated Date',
       'Community Board', 'Borough', 'X Coordinate (State Plane)',
       'Y Coordinate (State Plane)', 'Park Facility Name', 'Park Borough',
       'School Name', 'School Number', 'School Region', 'School Code',
       'School Phone Number', 'School Address', 'School City', 'School State',
       'School Zip', 'School Not Found', 'School or Citywide Complaint',
       'Vehicle Type', 'Taxi Company Borough', 'Taxi Pick Up Location',
       'Bridge Highway Name', 'Bridge Highway Direction', 'Road Ramp',
       'Bridge Highway Segment', 'Garage Lot Name', 'Ferry Direction',
       'Ferry Terminal Name', 'Latitude', 'Longitude', 'Location'],
      dtype='str')
```

In [6]:
```
print("Missing Values:\n", df.isnull().sum())
```

```
Missing Values:
 Unique Key                           0
Created Date                          0
Closed Date                        2381
Agency                                0
Agency Name                           0
Complaint Type                        0
Descriptor                         6501
Location Type                       133
Incident Zip                       2998
Incident Address                  51699
Street Name                       51699
Cross Street 1                    57188
Cross Street 2                    57805
Intersection Street 1            313438
Intersection Street 2            314046
Address Type                       3252
City                               2997
Landmark                         364183
Facility Type                      2389
Status                                0
Due Date                              3
Resolution Description                0
Resolution Action Updated Date     2402
Community Board                       0
Borough                               0
X Coordinate (State Plane)         4030
Y Coordinate (State Plane)         4030
Park Facility Name                    0
Park Borough                          0
School Name                           0
School Number                         0
School Region                         1
School Code                           1
School Phone Number                   0
School Address                        0
School City                           0
School State                          0
School Zip                            1
School Not Found                      0
School or Citywide Complaint     364558
Vehicle Type                     364558
Taxi Company Borough             364558
Taxi Pick Up Location            364558
Bridge Highway Name              364261
Bridge Highway Direction         364261
Road Ramp                        364296
Bridge Highway Segment           364296
Garage Lot Name                  364558
Ferry Direction                  364557
Ferry Terminal Name              364556
Latitude                           4030
Longitude                          4030
Location                           4030
dtype: int64
```

In [7]:
```
df.info()
```

```
<class 'pandas.DataFrame'>
RangeIndex: 364558 entries, 0 to 364557
Data columns (total 53 columns):
 #   Column                          Non-Null Count   Dtype
---  ------                          --------------   -----
 0   Unique Key                      364558 non-null  int64
 1   Created Date                    364558 non-null  str
 2   Closed Date                     362177 non-null  str
 3   Agency                          364558 non-null  str
 4   Agency Name                     364558 non-null  str
 5   Complaint Type                  364558 non-null  str
 6   Descriptor                      358057 non-null  str
 7   Location Type                   364425 non-null  str
 8   Incident Zip                    361560 non-null  float64
 9   Incident Address                312859 non-null  str
 10  Street Name                     312859 non-null  str
 11  Cross Street 1                  307370 non-null  str
 12  Cross Street 2                  306753 non-null  str
 13  Intersection Street 1           51120 non-null   str
 14  Intersection Street 2           50512 non-null   str
 15  Address Type                    361306 non-null  str
 16  City                            361561 non-null  str
 17  Landmark                        375 non-null     str
 18  Facility Type                   362169 non-null  str
 19  Status                          364558 non-null  str
 20  Due Date                        364555 non-null  str
 21  Resolution Description          364558 non-null  str
 22  Resolution Action Updated Date  362156 non-null  str
 23  Community Board                 364558 non-null  str
 24  Borough                         364558 non-null  str
 25  X Coordinate (State Plane)      360528 non-null  float64
 26  Y Coordinate (State Plane)      360528 non-null  float64
 27  Park Facility Name              364558 non-null  str
 28  Park Borough                    364558 non-null  str
 29  School Name                     364558 non-null  str
 30  School Number                   364558 non-null  str
 31  School Region                   364557 non-null  str
 32  School Code                     364557 non-null  str
 33  School Phone Number             364558 non-null  str
 34  School Address                  364558 non-null  str
 35  School City                     364558 non-null  str
 36  School State                    364558 non-null  str
 37  School Zip                      364557 non-null  str
 38  School Not Found                364558 non-null  str
 39  School or Citywide Complaint    0 non-null       float64
 40  Vehicle Type                    0 non-null       float64
 41  Taxi Company Borough            0 non-null       float64
 42  Taxi Pick Up Location           0 non-null       float64
 43  Bridge Highway Name             297 non-null     str
 44  Bridge Highway Direction        297 non-null     str
 45  Road Ramp                       262 non-null     str
 46  Bridge Highway Segment          262 non-null     str
 47  Garage Lot Name                 0 non-null       float64
 48  Ferry Direction                 1 non-null       str
 49  Ferry Terminal Name             2 non-null       str
 50  Latitude                        360528 non-null  float64
 51  Longitude                       360528 non-null  float64
 52  Location                        360528 non-null  str
dtypes: float64(10), int64(1), str(42)
memory usage: 147.4 MB
```

Dealing with duplicates

In [9]:
```
print("Duplicate rows:", df.duplicated().sum())
```

```
Duplicate rows: 0
```

Dealing with Null values

In [10]:
```
chunks = pd.read_csv(
    "311_Service_Requests_from_2010_to_Present.csv",
    chunksize=20000,
    low_memory=False
)

filtered_chunks = []

for chunk in chunks:
    chunk = chunk[chunk['Closed Date'].notnull()]
    filtered_chunks.append(chunk)

df = pd.concat(filtered_chunks, ignore_index=True)

df.head()
```

Out[10]:

	Unique Key	Created Date	Closed Date	Agency	Agency Name	Complaint Type	Descriptor
0	32310363	12/31/2015 11:59:45 PM	01/01/2016 12:55:15 AM	NYPD	New York City Police Department	Noise - Street/Sidewalk	Loud Music/Party
1	32309934	12/31/2015 11:59:44 PM	01/01/2016 01:26:57 AM	NYPD	New York City Police Department	Blocked Driveway	No Access
2	32309159	12/31/2015 11:59:29 PM	01/01/2016 04:51:03 AM	NYPD	New York City Police Department	Blocked Driveway	No Access
3	32305098	12/31/2015 11:57:46 PM	01/01/2016 07:43:13 AM	NYPD	New York City Police Department	Illegal Parking	Commercial Overnight Parking
4	32306529	12/31/2015 11:56:58 PM	01/01/2016 03:24:42 AM	NYPD	New York City Police Department	Illegal Parking	Blocked Sidewalk

5 rows × 53 columns

In [14]:
```
# Fill missing City
df['City'].fillna("Unknown City")
```

Out[14]:
```
0                 NEW YORK
1                  ASTORIA
2                    BRONX
3                    BRONX
4                 ELMHURST
                     ...
362172           WOODHAVEN
362173               BRONX
362174            NEW YORK
362175               BRONX
362176    SOUTH OZONE PARK
Name: City, Length: 362177, dtype: str
```

Exploring the data using visualization

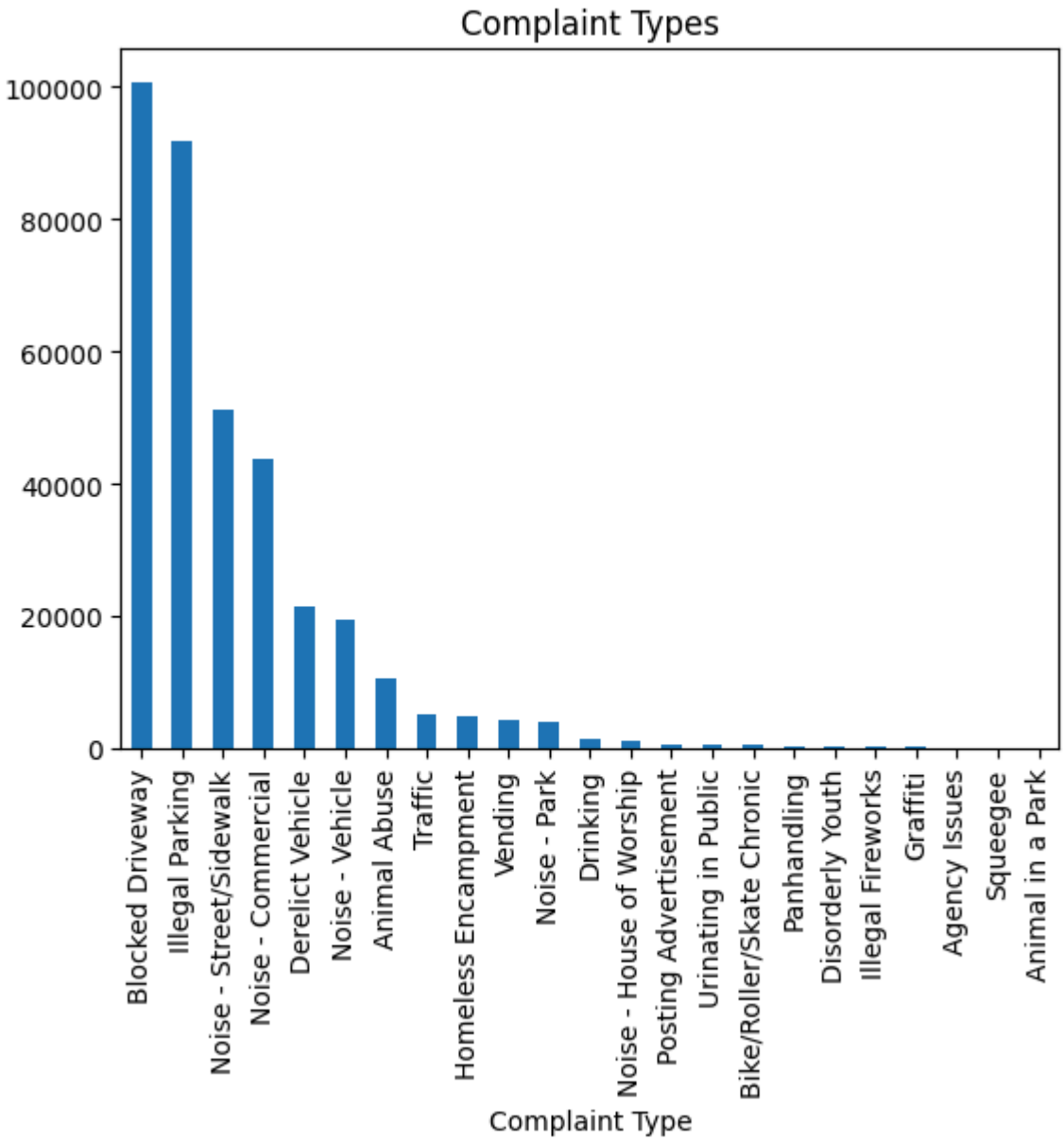
In [15]:
```
df['Request_Closing_Time'] = df['Closed Date'] - df['Created Date']
df['Request_Closing_Time'] = df['Request_Closing_Time'].dt.total_seconds()

print(df['Request_Closing_Time'].describe())
```

```
count    3.621770e+05
mean     1.511330e+04
std      2.110255e+04
min      6.100000e+01
25%      4.533000e+03
50%      9.616000e+03
75%      1.887800e+04
max      2.134342e+06
Name: Request_Closing_Time, dtype: float64
```

Complaint Type Analysis

In [16]:
```
complaint_counts = df['Complaint Type'].value_counts()

print("Top 10 Complaints:\n", complaint_counts.head(10))

complaint_counts.plot(kind='bar')
plt.title("Complaint Types")
plt.show()
```

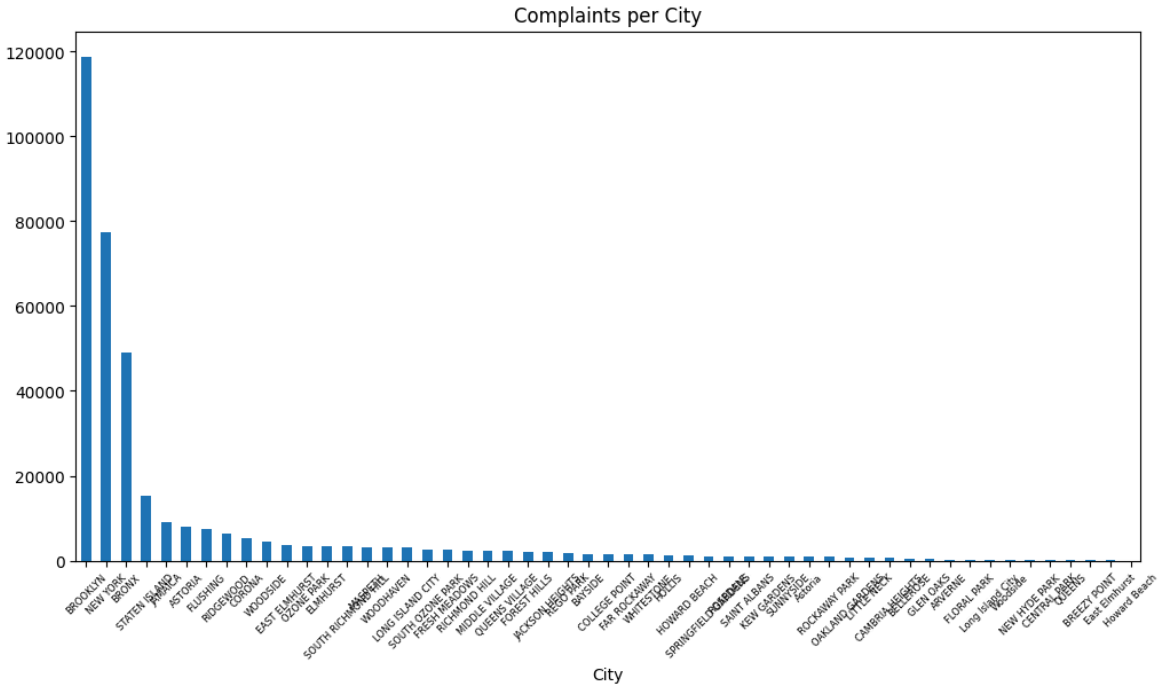
```
Top 10 Complaints:
 Complaint Type
Blocked Driveway           100624
Illegal Parking             91716
Noise - Street/Sidewalk     51139
Noise - Commercial          43751
Derelict Vehicle            21518
Noise - Vehicle             19301
Animal Abuse                10530
Traffic                      5196
Homeless Encampment          4879
Vending                      4185
Name: count, dtype: int64
```



City Analysis

In [31]:
```
city_counts = df['City'].value_counts()

print(city_counts)

city_counts.plot(kind='bar', figsize=(12,6))

plt.title("Complaints per City")
plt.xticks(fontsize=6, rotation=45)

plt.show()
```

```
City
BROOKLYN               118849
NEW YORK                77289
BRONX                   49166
STATEN ISLAND           15335
JAMAICA                  8930
ASTORIA                  7991
FLUSHING                 7486
RIDGEWOOD                6391
CORONA                   5383
WOODSIDE                 4357
EAST ELMHURST            3557
OZONE PARK               3446
ELMHURST                 3438
SOUTH RICHMOND HILL      3431
MASPETH                  3117
WOODHAVEN                3102
LONG ISLAND CITY         3027
SOUTH OZONE PARK         2668
FRESH MEADOWS            2453
RICHMOND HILL            2333
MIDDLE VILLAGE           2291
QUEENS VILLAGE           2251
FOREST HILLS             2122
JACKSON HEIGHTS          2105
REGO PARK                1807
BAYSIDE                  1550
COLLEGE POINT            1544
FAR ROCKAWAY             1397
WHITESTONE               1369
HOLLIS                   1231
HOWARD BEACH             1144
SPRINGFIELD GARDENS      1094
ROSEDALE                 1091
SAINT ALBANS             1047
KEW GARDENS              1008
SUNNYSIDE                 944
Astoria                   905
ROCKAWAY PARK             831
OAKLAND GARDENS           717
LITTLE NECK               712
CAMBRIA HEIGHTS           617
BELLEROSE                 487
GLEN OAKS                 361
ARVERNE                   259
FLORAL PARK               196
Long Island City          170
Woodside                  166
NEW HYDE PARK             129
CENTRAL PARK              110
QUEENS                     37
BREEZY POINT               31
East Elmhurst              30
Howard Beach                1
Name: count, dtype: int64
```



Complaint Type vs City Table

In [22]:
```
df_new = pd.crosstab(df['Complaint Type'], df['City'])
print(df_new)
```

City Complaint Type	ARVERNE	ASTORIA	Astoria	BAYSIDE	BELLEROSE	\
Animal Abuse	46	170	0	53	15	
Animal in a Park	0	0	0	0	0	
Bike/Roller/Skate Chronic	0	16	0	0	1	
Blocked Driveway	50	3436	159	514	138	
Derelict Vehicle	32	426	14	231	120	
Disorderly Youth	2	5	0	2	2	
Drinking	1	43	0	1	1	
Graffiti	1	4	0	3	0	
Homeless Encampment	4	32	0	2	1	
Illegal Fireworks	0	4	0	0	1	
Illegal Parking	62	1340	277	638	132	
Noise - Commercial	2	1653	310	47	38	
Noise - House of Worship	14	21	0	3	1	
Noise - Park	2	64	0	4	1	
Noise - Street/Sidewalk	29	409	145	17	13	
Noise - Vehicle	10	236	0	24	11	
Panhandling	1	2	0	0	1	
Posting Advertisement	0	3	0	0	1	
Squeegee	0	0	0	0	0	
Traffic	1	60	0	9	9	
Urinating in Public	1	10	0	0	1	
Vending	1	57	0	2	0	

City Complaint Type	BREEZY POINT	BRONX	BROOKLYN	CAMBRIA HEIGHTS	\
Animal Abuse	2	1971	3191	15	
Animal in a Park	0	0	0	0	
Bike/Roller/Skate Chronic	0	22	124	0	
Blocked Driveway	3	17062	36445	177	
Derelict Vehicle	3	2402	6257	148	
Disorderly Youth	0	66	79	0	
Drinking	1	206	291	0	
Graffiti	0	15	60	0	
Homeless Encampment	0	275	948	6	
Illegal Fireworks	0	24	61	1	
Illegal Parking	16	9889	33532	113	
Noise - Commercial	4	2944	13855	19	
Noise - House of Worship	0	90	389	2	
Noise - Park	0	548	1575	0	
Noise - Street/Sidewalk	1	9144	13982	29	
Noise - Vehicle	1	3556	5965	100	
Panhandling	0	20	49	0	
Posting Advertisement	0	18	58	0	
Squeegee	0	0	0	0	
Traffic	0	427	1258	7	
Urinating in Public	0	54	155	0	
Vending	0	433	575	0	

City Complaint Type	CENTRAL PARK	... ...	SAINT ALBANS	SOUTH OZONE PARK	\
Animal Abuse	0	...	43	74	
Animal in a Park	0	...	0	0	
Bike/Roller/Skate Chronic	0	...	0	1	
Blocked Driveway	0	...	318	1202	
Derelict Vehicle	0	...	248	425	
Disorderly Youth	0	...	1	2	
Drinking	0	...	3	14	
Graffiti	0	...	0	2	

Homeless Encampment	0	...	11	5
Illegal Fireworks	0	...	0	1
Illegal Parking	5	...	237	602
Noise - Commercial	0	...	36	82
Noise - House of Worship	0	...	1	5
Noise - Park	0	...	1	4
Noise - Street/Sidewalk	105	...	81	108
Noise - Vehicle	0	...	50	97
Panhandling	0	...	0	0
Posting Advertisement	0	...	0	1
Squeegee	0	...	0	0
Traffic	0	...	14	36
Urinating in Public	0	...	1	2
Vending	0	...	2	5

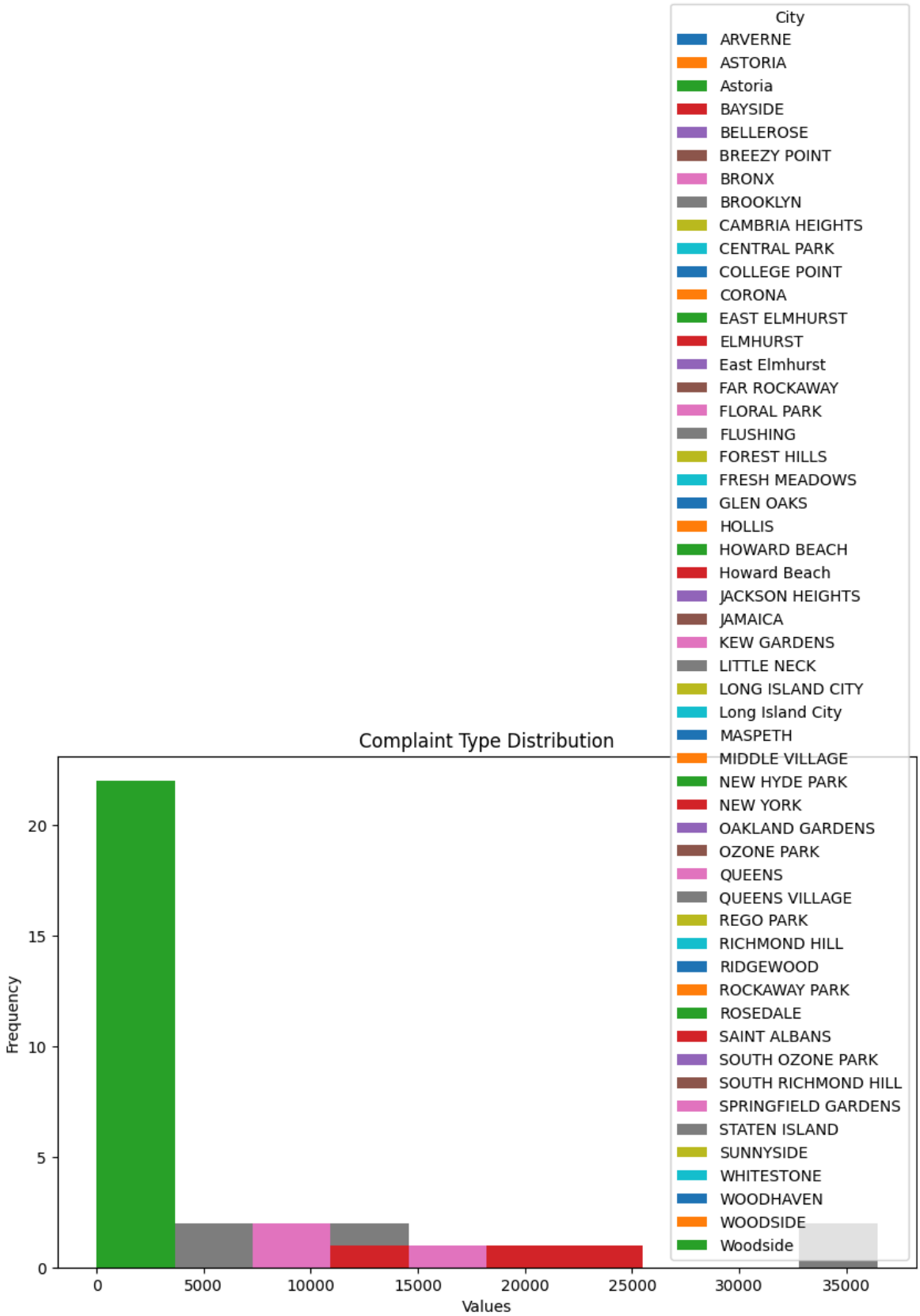
City	SOUTH RICHMOND HILL	SPRINGFIELD GARDENS	\
Complaint Type			
Animal Abuse	40	42	
Animal in a Park	0	0	
Bike/Roller/Skate Chronic	1	0	
Blocked Driveway	1946	330	
Derelict Vehicle	356	267	
Disorderly Youth	2	0	
Drinking	25	6	
Graffiti	0	0	
Homeless Encampment	12	7	
Illegal Fireworks	2	1	
Illegal Parking	596	291	
Noise - Commercial	223	38	
Noise - House of Worship	3	1	
Noise - Park	2	1	
Noise - Street/Sidewalk	93	42	
Noise - Vehicle	93	48	
Panhandling	0	2	
Posting Advertisement	0	2	
Squeegee	0	0	
Traffic	12	12	
Urinating in Public	1	3	
Vending	24	1	

City	STATEN ISLAND	SUNNYSIDE	WHITESTONE	WOODHAVEN	\
Complaint Type					
Animal Abuse	786	40	43	57	
Animal in a Park	0	0	0	0	
Bike/Roller/Skate Chronic	10	2	4	2	
Blocked Driveway	2845	278	279	1363	
Derelict Vehicle	2184	17	279	369	
Disorderly Youth	25	2	1	0	
Drinking	188	12	3	4	
Graffiti	6	1	1	0	
Homeless Encampment	77	12	0	10	
Illegal Fireworks	11	0	1	0	
Illegal Parking	6224	167	631	896	
Noise - Commercial	783	238	21	209	
Noise - House of Worship	18	0	0	3	
Noise - Park	67	16	7	3	
Noise - Street/Sidewalk	885	69	35	89	
Noise - Vehicle	424	53	31	81	
Panhandling	13	0	0	1	
Posting Advertisement	516	3	0	0	

```
Squeegee                                0          0          0          0
Traffic                               229         17         32          7
Urinating in Public                    19          2          0          2
Vending                                25         15          1          6

City                        WOODSIDE  Woodside
Complaint Type
Animal Abuse                     111         0
Animal in a Park                   0         0
Bike/Roller/Skate Chronic          5         0
Blocked Driveway                2038        27
Derelict Vehicle                 298         8
Disorderly Youth                   1         0
Drinking                          15         0
Graffiti                           4         0
Homeless Encampment               38         0
Illegal Fireworks                  1         0
Illegal Parking                 1083       124
Noise - Commercial               256         2
Noise - House of Worship           4         0
Noise - Park                      38         0
Noise - Street/Sidewalk          261         5
Noise - Vehicle                  136         0
Panhandling                        0         0
Posting Advertisement              0         0
Squeegee                           0         0
Traffic                           45         0
Urinating in Public                8         0
Vending                           15         0

[22 rows x 53 columns]
```

histrogram Bar Chart

In [30]:
```
df_new.plot(kind='hist', figsize=(10,6))
plt.title("Complaint Type Distribution")
plt.xlabel("Values")
plt.ylabel("Frequency")
plt.show()
```

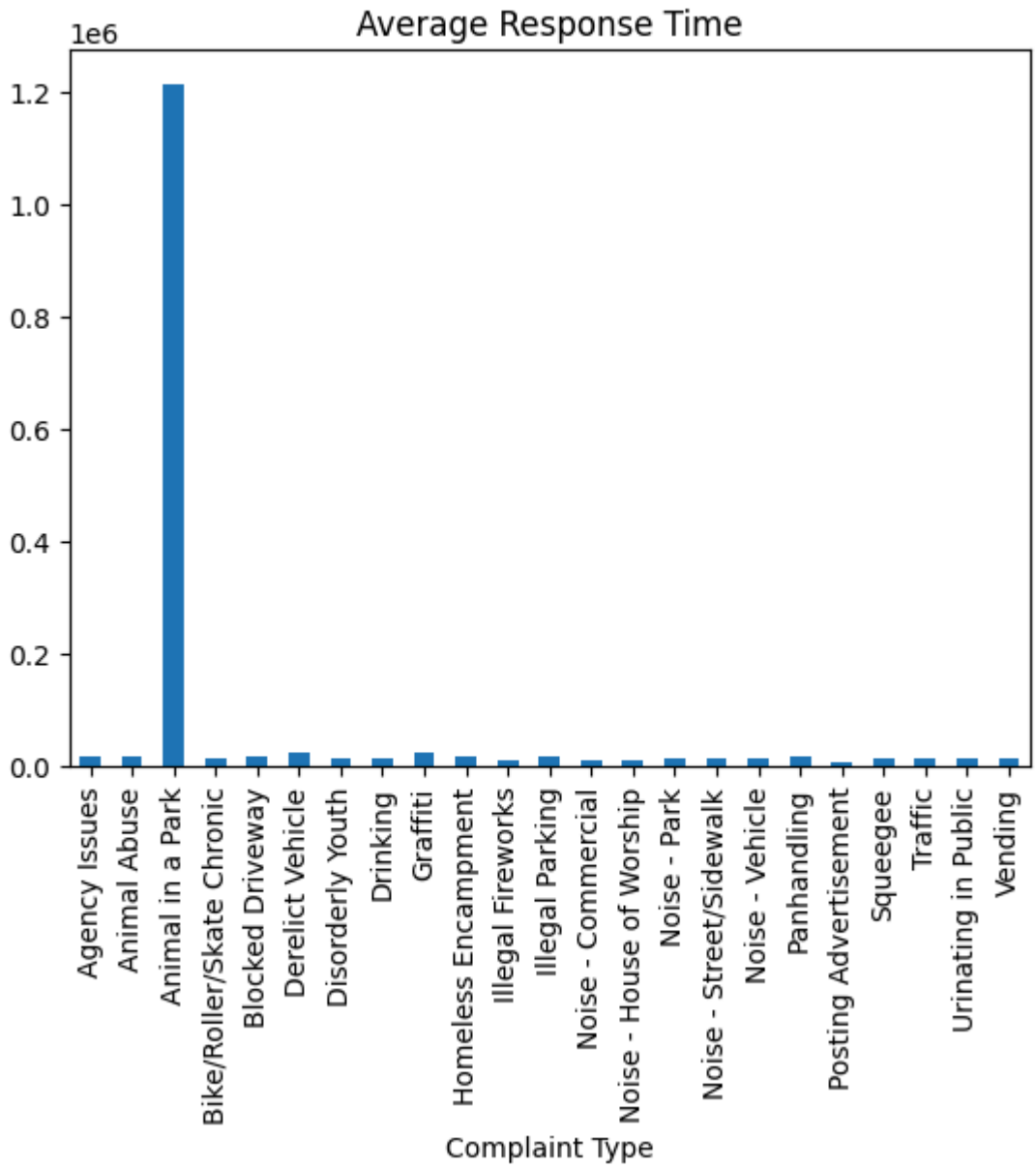


Response Time Analysis

In [25]:
```
avg_time = df.groupby('Complaint Type')['Request_Closing_Time'].mean()

print(avg_time)

avg_time.plot(kind='bar')
```

```
plt.title("Average Response Time")
plt.show()
```

```
Complaint Type
Agency Issues                1.828912e+04
Animal Abuse                 1.803256e+04
Animal in a Park             1.212634e+06
Bike/Roller/Skate Chronic    1.312369e+04
Blocked Driveway             1.623252e+04
Derelict Vehicle             2.535960e+04
Disorderly Youth             1.236375e+04
Drinking                     1.382130e+04
Graffiti                     2.327634e+04
Homeless Encampment          1.545138e+04
Illegal Fireworks            1.011348e+04
Illegal Parking              1.565044e+04
Noise - Commercial           1.108576e+04
Noise - House of Worship     1.139109e+04
Noise - Park                 1.222606e+04
Noise - Street/Sidewalk      1.223130e+04
Noise - Vehicle              1.256180e+04
Panhandling                  1.585355e+04
Posting Advertisement        7.286256e+03
Squeegee                     1.456025e+04
Traffic                      1.230912e+04
Urinating in Public          1.295929e+04
Vending                      1.436628e+04
Name: Request_Closing_Time, dtype: float64
```



Kruskal-Wallis Test

In [26]:
```
groups = df.groupby('Complaint Type')['Request_Closing_Time'].apply(list)

stat, p = kruskal(*groups)

print("P-value:", p)

if p < 0.05:
    print("Reject Null Hypothesis (Different response times)")
else:
    print("Fail to Reject Null Hypothesis (Similar response times)")
```

```
P-value: 0.0
Reject Null Hypothesis (Different response times)
```

In [ ]: